On the Capacity of the Discrete Memoryless Broadcast Channel with Feedback

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***Abstract*—A coding scheme for the discrete memoryless broadcast channel with {noiseless, noisy, generalized} feedback is proposed, and the associated achievable region derived. The scheme is based on a block-Markov strategy combining the Marton scheme and a lossy version of the Gray-Wyner scheme with side-information, where in each block the transmitter sends fresh data and update information that allows the receivers to improve the channel outputs observed in the previous block. The region is evaluated for two specific broadcast channels : A generalization of Dueck's channel, and a noisy version of Blackwell's channel. With noiseless feedback, the region is shown to exceed the corresponding no-feedback capacity region in both cases, and for the former channel also yields the noiseless-feedback capacity region. A simple continuity argument verifies an improvement over the no-feedback capacity region for both channels, in the limit of low feedback noise.**

I. Introduction

We consider a broadcast channel (BC) with two receivers, where the transmitter has instantaneous access to a feedback signal. Popular examples of such feedback signals are:

- the channel outputs observed at the two receivers (this setup is called *noiseless feedback*); or
- a noisy version of these channel outputs (this setup is called *noisy feedback*).

Here we allow for very general feedback signals, and only require that the time-t feedback signal is obtained by feeding the time-t input and the corresponding time-t outputs into a memoryless feedback channel. This general form of feedback is commonly referred to as *generalized feedback*. For brevity, here we mostly omit the word *generalized*. It is easily seen that our setup includes noiseless feedback and noisy feedback as special cases.

We focus on discrete memoryless broadcast channels (DMBCs), namely where the input and output symbols are from finite alphabets and where the time-t channel outputs depend on the past inputs and outputs only through the time-t input. Our interest lies in the feedback-capacity region of such DMBCs, i.e., in the associated set of rate tuples for which reliable communication is possible.

Most previous results on BCs with feedback focus on the case of noiseless feedback. For example, El Gamal [1] proved that when the BC is physically degraded, i.e., one of the two outputs is obtained by processing the other output, then the capacity region with noiseless-feedback coincides with the no-feedback capacity region. In contrast, Dueck [2] and Kramer [3] described some specific examples of DMBCs where the noiseless-feedback capacity region exceeds the no-feedback capacity region, In Dueck's example, the noiseless-feedback capacity region is known. However, outside these specific examples, determining the capacity region with feedback for (non-physically-degraded) BCs is an open problem. In fact, even characterizing the class of DMBCs where feedback enlarges the capacity region seems hard. This is partly because even the no-feedback capacity region is generally unknown, and partly because a computable single-letter achievable region for the DMBC with feedback was missing hitherto. Kramer [3] proposed a multi-letter achievable region for the DMBC with noisy or noiseless feedback.

In this paper we propose a coding scheme for the DMBC with generalized feedback, and present a corresponding single-letter achievable region. Subsequently, we analyze two new examples – a generalization of Dueck's channel [2], and a noisy version of Blackwell's channel [4] – where our region is shown to exceed the no-feedback capacity region, even in the presence of feedback noise.[1] Our approach is motivated by Dueck's example [2], and is based on the following idea. The transmitter uses the feedback to identify update information that is useful to the receivers when decoding their intended messages, and describes this information in subsequent transmissions. More specifically, our scheme adopts a block-Markov strategy, where in each block the transmitter sends a combination of fresh data and compressed update information pertaining to the data sent in the previous block. Marton's no-feedback scheme [7], [8] is used in each block to send the fresh data and the update information, at rates outside the no-feedback capacity. The update information sent in a block is essentially an efficient lossy description of the auxiliary inputs in Marton's scheme from the previous block, taking into account the receivers' observations and the feedback signal as side-information. The receivers perform backward decoding; starting with the last block, each receiver iteratively performs the following two steps: 1) it decodes its intended data and update information in the current block; and 2) it uses the update information to "improve" the channel outputs in the preceding block, which is processed next. Loosely speaking, this strategy is gainful whenever the cost of the lossy description is smaller than the increase in rate is supports.

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[1] An earlier version of this result was presented by the authors at ISIT 2010 [5]. In the same session, another achievable region for the DMBC, in the noiseless feedback case, was presented [6] and shown to coincide with Dueck's example in his original setting. However, this latter region seems difficult to evaluate in general. Whether it yields an improvement over the no-feedback capacity in any other case, or how it is related to our region, currently seems unclear.

Intuitively, this is expected to happen when the descriptions required by the two receivers have a large common part.

The paper is organized as follows. In Section II, the necessary mathematical background is provided. The channel model in described in Section III. In Section IV, Marton's scheme for the DMBC without feedback is reviewed in detail. In Section V, a lossy version with side-information of the Gray-Wyner distributed source coding setup is introduced, and an achievable region is obtained. The main result of the paper is introduced In Section VI, where the Marton and the lossy Gray-Wyner schemes are combined into a feedback scheme for general DMBCs, and the associated achievable region is derived. Two new examples are discussed in VII: A generalization of Dueck's DMBC , and a noisy version of Blackwell's DMBC [4]. In both cases, the the region achieved by the new scheme is shown to exceed the no-feedback capacity region, using either noiseless feedback or noisy feedback, in the limit of low feedback noise.

II. Preliminaries

A. Notations

For any real number $M > 1$, we use the notation $[M] \stackrel{\text{def}}{=} \{1, \ldots, \lfloor M \rfloor\}$. We think of a product set of the form $[2^{nr_1}] \times [2^{nr_2}]$ as being one-to-one with $[2^{n(r_1+r_2)}]$, disregarding the associated integer issues throughout. For $\epsilon > 0$, we write $\delta(\epsilon)$ to indicate a general nonnegative function satisfying $\delta(\epsilon) \to 0$ (arbitrarily slow) as $\epsilon \to 0$.

A random sequence $X^n \stackrel{\text{def}}{=} (X_1, \ldots, X_n)$ is said to be P_X-*independent-identically distributed (P_X-i.i.d.)* if

$$P_{X^n}(x^n) = \prod_{t=1}^{n} P_X(x_t)$$

for all x^n. Let (X^n, Y^n) be two jointly distributed random sequences, and let $P_{Y|X}$ be some conditional distribution. We say that Y^n is $P_{Y|X}$-*independent given* X^n if

$$P_{Y^n|X^n}(y^n|x^n) = \prod_{t=1}^{n} P_{Y|X}(y_t|x_t)$$

for all y^n and x^n with $P_{X^n}(x^n) > 0$.

We use the notion of typicality as defined in [9]. For a finite alphabet $\mathcal{X}$, a sequence $x^{\in}\mathcal{X}^n$ is said to be ϵ-typical with respect to (w.r.t.) a distribution P_X on $\mathcal{X}$ if

$$|\pi_{x^n}(x) - P_X(x)| \leq \epsilon \cdot P_X(x)$$

for all $x \in \mathcal{X}$, where π_{x^n} is the distribution over $\mathcal{X}$ corresponding to the relative frequency of symbols in x^n. The set of all such sequences is denoted $\mathcal{T}_\epsilon^n(P_X)$. Similarly, for a law $P_{X_1 \cdots X_k}$ over a product alphabet $\mathcal{X}_1 \times \cdots \times \mathcal{X}_k$, we denote by $\mathcal{T}_\epsilon^n(P_{X_1 \cdots X_k})$ the set of all k-tuples of sequences $(x_1^n \in \mathcal{X}_1^n, \ldots, x_k^n \in \mathcal{X}_k^n)$ that are *jointly* ϵ-typical w.r.t. $P_{X_1 \cdots X_k}$.

B. Basic Lemmas

The following three lemmas are well known, and used extensively in the sequel.

Lemma 1 (Conditional Typicality Lemma [9])**.** *Let P_{XY} be some joint distribution. Suppose $x^n \in \mathcal{T}_{\epsilon'}^n(P_X)$ for some $\epsilon' > 0$, and Y^n is $P_{Y|X}$-independent given $X^n = x^n$. Then for every $\epsilon > \epsilon'$:*

$$\lim_{n\to\infty} \Pr\left((x^n, Y^n) \notin \mathcal{T}_\epsilon^n(P_{XY})\right) = 0.$$

Lemma 2 (Covering Lemma [9])**.** *Let $0 < \epsilon' < \epsilon$. Let X^n satisfy $\Pr(X^n \in \mathcal{T}_{\epsilon'}(P_X)) \to 1$ as $n \to \infty$, and let $\{Y^n(m)\}_{m=1}^M$ be a set of P_Y-i.i.d. sequences such that $\{X^n, \{Y^n(m)\}_{m=1}^M\}$ are mutually independent and $M \geq 2^{nr}$. Then there exists $\delta(\epsilon) \to 0$ as $\epsilon \to 0$ such that*

$$\lim_{n\to\infty} \Pr\left(\forall m \in [M],\ (X^n, Y^n(m)) \notin \mathcal{T}_\epsilon^n(P_{XY})\right) = 0$$

if $r > I(X;Y) + \delta(\epsilon)$.

Lemma 3 (Packing Lemma [9])**.** *Let $\epsilon > 0$. Let P_{XY} be some joint distribution. Let X^n be an arbitrarily distributed sequence. Let $\{Y^n(m)\}_{m=1}^M$ be a set of P_Y-i.i.d. random sequences, where $Y^n(m)$ is independent of X^n and $M \leq 2^{nr}$. Then there exists $\delta(\epsilon) \to 0$ as $\epsilon \to 0$ such that*

$$\lim_{n\to\infty} \Pr\left(\exists m \in [M] \quad s.t. \quad (X^n, Y^n(m)) \in \mathcal{T}_\epsilon^n(P_{XY})\right) = 0$$

if $r < I(X;Y) - \delta(\epsilon)$.

The following is a simple multivariate generalization of the packing Lemma.

Lemma 4 (Multivariate packing Lemma)**.** *Let $\epsilon > 0$. Let $(U_1, U_2, U_3) \sim P_{U_1U_2U_3}$. For $i \in \{1,2,3\}$, let $\{U_i^n(m)\}_{m=1}^{M_i}$ be three sets of P_{U_i}-i.i.d. random vectors, respectively, where $M_i \leq 2^{nr_i}$. Assume that $\{U_1^n(m_1), U_2^n(m_2), U_3^n(m_3)\}$ are mutually independen for any m_1, m_2, m_3. Then there exists $\delta(\epsilon) \to 0$ as $\epsilon \to 0$ such that*

$$\lim_{n\to\infty} \Pr(\exists\, m_i \in [M_i] \ \ for \ \ i \in \{1,2,3\} \quad s.t. \\ (U_1^n(m_1), U_2^n(m_2), U_3^n(m_3)) \in T_\epsilon^n(P_{U_1U_2U_3})) = 0$$

if $r_1 + r_2 + r_3 < I(U_1;U_2) + I(U_3;U_1,U_2) - \delta(\epsilon)$.

Proof outline. Let $\mathcal{E}_{ijk} \stackrel{\text{def}}{=} \{U_1^n(i), U_2^n(j), U_3^n(k)) \in T_\epsilon^n(P_{U_1U_2U_3})\}$. We need to show that $\Pr\left(\bigcup_{ijk} \mathcal{E}_{ijk}\right) \to 0$ under the constraint. By standard typicality/large deviation arguments we have that

$$\begin{aligned}\Pr(\mathcal{E}_{ijk}) &\leq 2^{-n\left(D(P_{U_1U_2U_3}\|P_{U_1}\times P_{U_3}\times P_{U_3})-\delta(\epsilon)\right)} \\ &= 2^{-n\left(D(P_{U_1U_2U_3}\|P_{U_1}\times P_{U_3}\times P_{U_3})-\delta(\epsilon)\right)} \\ &= 2^{-n(D(I(U_1;U_2)+I(U_3;U_1,U_2)-\delta(\epsilon))}\end{aligned}$$

The result follows by taking the union bound over $\mathcal{E}_{ijk}$, and requiring that it tends to zero. ■

III. Channel Model

We describe the channel model in more detail. The goal of the communication is that the transmitter conveys a private Message M_1 to a Receiver 1, a private Message M_2 to a Receiver 2, and a common message M_0 to both receivers. The three messages M_0, M_1, and M_2 are assumed to be independent and uniformly distributed over the finite sets $[2^{nR_0}]$, $[2^{nR_1}]$, and $[2^{nR_2}]$ respectively, where n denotes the

blocklength and R_0, R_1, R_2 are the corresponding common and private transmission rates.

Communication takes place over a DMBC with generalized feedback. This channel is characterized by a quadruple of finite alphabets $\mathcal{X}$, $\mathcal{Y}_1, \mathcal{Y}_2$, and $\widetilde{\mathcal{Y}}$, and a conditional probability law $P_{Y_1Y_2\widetilde{Y}|X}(y_1, y_2, \widetilde{y}|x)$ where $x \in \mathcal{X}$, $y_1 \in \mathcal{Y}_1$, $y_2 \in \mathcal{Y}_2$, and $\widetilde{y} \in \widetilde{\mathcal{Y}}$. Given that at time t the transmitter feeds the symbol x_t to the channel, then Receiver 1 and Receiver 2 observe the channel outputs $y_{1,t} \in \mathcal{Y}_1$ and $y_{2,t} \in \mathcal{Y}_2$ respectively and the transmitter observes the generalized feedback $\widetilde{y}_t \in \widetilde{\mathcal{Y}}$, with probability $P_{Y_1Y_2\widetilde{Y}|X}(y_{1,t}, y_{2,t}, \widetilde{y}_t|x_t)$.

Thanks to feedback, the transmitter can produce its time-t channel input X_t as a function of the Messages M_0, M_1, M_2 and of the previously observed feedback outputs $\widetilde{Y}^{t-1} \stackrel{\text{def}}{=} (\widetilde{Y}_1, \ldots, \widetilde{Y}_{t-1})$:

$$X_t = \psi_t^{(n)}\left(M_0, M_1, M_2, \widetilde{Y}^{t-1}\right). \tag{1}$$

The DMBC and its feedback channel are memoryless which is captured by the following Markov relation for $t \in [n]$:

$$(Y_1^{t-1}, Y_2^{t-1}, \widetilde{Y}^{t-1}) \multimap X_t \multimap (Y_{1,t}, Y_{2,t}, \widetilde{Y}_t)$$

where $Y_i^{t-1} \stackrel{\text{def}}{=} (Y_{i,1}, Y_{i,2}, \ldots, Y_{i,t-1})$, for $i \in \{1, 2\}$.

After n channel uses Receiver i decodes its intended messages M_0 and M_i for $i \in \{1, 2\}$. Namely, Receiver i produces the guess:

$$(\hat{M}_{0,i}, \hat{M}_i) = \Psi_i^{(n)}(Y_i^n), \quad i \in \{1, 2\}. \tag{2}$$

A rate triplet (R_0, R_1, R_2) is called achievable if for every blocklength n there exists a set of n encoding functions $\left\{\psi_t^{(n)}\right\}_{t=1}^n$ and two decoding functions $\Psi_1^{(n)}$ and $\Psi_2^{(n)}$ such that the probability of decoding error, i.e., the probability that

$$(M_0, M_1) \neq (\hat{M}_{0,1}, \hat{M}_1) \quad \text{or} \quad (M_0, M_2) \neq (\hat{M}_{0,2}, \hat{M}_2),$$

tends to 0 as the blocklength n tends to infinity. The closure of the set of achievable rate triplets (R_0, R_1, R_2) is called the *feedback capacity-region* of this setup, and we denote it by $\mathcal{C}_{\text{GenFB}}$. The supremum of the sum $R_1 + R_2$ over all achievable rate triplets is called its *feedback sum-capacity*, and we denote it by $C_{\text{GenFB},\Sigma}$.

The described generalized-feedback setup includes as special cases the *no-feedback* setup where the feedback outputs are deterministic, e.g., $|\widetilde{\mathcal{Y}}| = 1$; the *noiseless-feedback* setup where the feedback output coincides with the pair of channel outputs, i.e., $\widetilde{Y} = (Y_1, Y_2)$; and the *noisy-feedback* setup where the feedback outputs and the channel inputs and outputs satisfy the Markov relation $X_t \multimap (Y_{1,t}, Y_{2,t}) \multimap \widetilde{Y}_t$ for all $t \in [n]$. In these special cases, we denote the capacity regions by $\mathcal{C}_{\text{NoFB}}$, $\mathcal{C}_{\text{NoiselessFB}}$ and $\mathcal{C}_{\text{NoisyFB}}$ respectively, and the corresponding sum-capacities by $C_{\text{NoFB},\Sigma}$, $C_{\text{NoiselessFB},\Sigma}$ and $C_{\text{NoisyFB},\Sigma}$ respectively.

IV. Marton's No-Feedback Scheme

In this section, we review the Marton coding scheme for the DMBC with a common message [7], [8]. This detailed description is necessary for the embedding of this scheme in our construction in Section VI.

A. Achievable Region

Let $\mathcal{R}_{\text{Marton}}$ be the closure of the set of all nonnegative rate triplets (R_0, R_1, R_2) that for some choice of random variables U_0, U_1, U_2 over finite alphabets $\mathcal{U}_0, \mathcal{U}_1, \mathcal{U}_2$ and some function $f\colon \mathcal{U}_0 \times \mathcal{U}_1 \times \mathcal{U}_2 \to \mathcal{X}$ satisfy

$$\begin{aligned} R_0 &< \min_i I(U_0; Y_i) \\ R_0 + R_1 &< I(U_0, U_1; Y_1) \\ R_0 + R_2 &< I(U_0, U_2; Y_2) \\ R_0 + R_1 + R_2 &< I(U_1; Y_1|U_0) + I(U_2; Y_2|U_0) \\ &\quad + \min_i I(U_0; Y_i) - I(U_1; U_2|U_0) \end{aligned} \tag{3}$$

where $X = f(U_0, U_1, U_2)$ and $(Y_1, Y_2) \sim P_{Y_1Y_2|X}$ given X.

Notice that we thus allow all triplets U_0, U_1, U_2 satisfying the Markov chain $(U_0, U_1, U_2) \multimap X \multimap (Y_1, Y_2)$.

Theorem 1 (From [7], [8]). $\mathcal{R}_{\text{Marton}} \subseteq \mathcal{C}_{\text{NoFB}}$.

In the following subsection we recall a scheme achieving this region and sketch its analysis.

B. Marton's Scheme

We describe the scheme for a DMBC $(\mathcal{X}, \mathcal{Y}_1, \mathcal{Y}_2, P_{Y_1Y_2|X})$. The scheme has parameters $(\mathcal{U}_0, \mathcal{U}_1, \mathcal{U}_2, P_{U_0U_1U_2}, f, R_0, R_{1,p}, R_{1,c}, R_{2,p}, R_{2,c}, R_1', R_2', \epsilon, n)$ where

- $\mathcal{U}_0, \mathcal{U}_1, \mathcal{U}_2$ are auxiliary finite alphabets;
- $P_{U_0U_1U_2}$ is a joint law over these auxiliary alphabets;
- $f\colon \mathcal{U}_0 \times \mathcal{U}_1 \times \mathcal{U}_2 \to \mathcal{X}$ is a function mapping the auxiliary inputs into effective inputs;
- $R_0, R_{1,p}, R_{2,p}, R_{1,c}, R_{2,c}$ are nonnegative communication rates;
- R_1', R_2' are nonnegative binning rates;
- $\epsilon > 0$ is a small number; and
- n denotes the scheme's blocklength.

As we shall see, the scheme uses a rate-splitting approach: each private message M_i, for $i \in \{1, 2\}$, is split into two independent submessages $M_{i,p}$ and $M_{i,c}$, where $M_{i,p}$ is uniformly distributed over $[2^{nR_{1,p}}]$ and $M_{i,c}$ is uniform over $[2^{nR_{i,c}}]$. Thus, in our scheme, the private messages M_1 and M_2 are of total rates $R_1 \stackrel{\text{def}}{=} R_{1,p} + R_{1,c}$ and $R_2 \stackrel{\text{def}}{=} R_{2,p} + R_{2,c}$. The submessage $M_{i,c}$ will be decoded by both receivers, and hence we call it the common part of Message M_i. The submessage $M_{i,p}$ is only decoded by its intended receiver i, and we call it the private part of M_i.

As we shall see, in our scheme Message M_0 is of rate R_0. Thus, the total rate of the messages decoded by both receivers equals $R_c \stackrel{\text{def}}{=} R_0 + R_{1,c} + R_{2,c}$. We denote the associated product common message by $M_c \stackrel{\text{def}}{=} (M_0, M_{1,c}, M_{2,c})$.

1) Code Construction: Our code consists of a single codebook $\mathcal{C}_0$, of $\lfloor 2^{nR_c} \rfloor$ codebooks $\{\mathcal{C}_1(m_0)\}_{m_0=1}^{\lfloor 2^{nR_c} \rfloor}$, and of $\lfloor 2^{nR_c} \rfloor$ codebooks $\{\mathcal{C}_2(m_0)\}_{m_0=1}^{\lfloor 2^{nR_c} \rfloor}$.

Codebook $\mathcal{C}_0$ consists of $\lfloor 2^{nR_c} \rfloor$ codewords of length n. Each codeword in $\mathcal{C}_0$ is constructed by randomly and independently drawing all its entries according to the distribution P_{U_0}. We denote the ℓ-th codeword in $\mathcal{C}_0$ by $u_0^n(\ell)$.

Every codebook $\{\mathcal{C}_1(m_0)\}$ consists of $\lfloor 2^{nR_{1,p}} \rfloor$ bins, each bin containing $\lfloor 2^{nR_1'} \rfloor$ codewords of length n. We denote the codewords in bin $m_1 \in [2^{nR_{1,p}}]$ by $\{u_1^n(m_0, m_1, \ell_1)\}_{\ell_1=1}^{\lfloor 2^{nR_1'} \rfloor}$. Each codeword $u_1^n(m_0, m_1, \ell_1)$ in $\mathcal{C}_1(m_0)$ is constructed by randomly drawing its j-th entry independent of all other entries in the codebook and according to the law $P_{U_1|U_0}(\cdot|u_{0,j}(m_0))$, where $u_{0,j}(m_0)$ denotes the j-th entry of $u_0^n(m_0)$.

Similarly, every codebook $\{\mathcal{C}_2(m_0)\}$ consists of $\lfloor 2^{nR_{2,p}} \rfloor$ bins, each bin containing $\lfloor 2^{nR_2'} \rfloor$ codewords of length n. We denote the codewords in bin $m_2 \in [2^{nR_{2,p}}]$ by $\{u_2^n(m_0, m_2, \ell_2)\}_{\ell_2=1}^{\lfloor 2^{nR_2'} \rfloor}$. Each codeword $u_2^n(m_0, m_2, \ell_2)$ in $\mathcal{C}_2(m_0)$ is constructed by randomly drawing its j-th entry independent of all other entries in the codebook and according to the law $P_{U_2|U_0}(\cdot|u_{0,j}(m_0))$.

Reveal all codebooks to the transmitter; codebooks $\mathcal{C}_0$ and $\{\mathcal{C}_1(m_0)\}$ to Receiver 1, and codebooks $\mathcal{C}_0$ and $\{\mathcal{C}_2(m_0)\}$ to Receiver 2.

2) Encoding: As previously described, the encoder first parses both private messages $M_1 \in [2^{nR_1}]$ and $M_2 \in [2^{nR_2}]$ into two independent submessages: $(M_{1,p}, M_{1,c})$ and $(M_{2,p}, M_{2,c})$. Then, with the Messages $M_0, M_{1,c}, M_{2,c}$ intended for both receivers it forms the new common message M_c.

The encoding takes place as follows. Given that $M_c = m_0$, $M_{1,p} = m_1$, $M_{2,p} = m_2$, the encoder searches the codebooks $\mathcal{C}_1(m_0)$ and $\mathcal{C}_2(m_0)$ for pairs of codewords $u_1^n(m_0, m_1, \ell_1), u_2^n(m_0, m_2, \ell_2)$ that satisfy[2]

$$(u_0^n(m_0), u_1^n(m_0, m_1, \ell_1), u_2^n(m_0, m_2, \ell_2)) \in \mathcal{T}_{\epsilon/32}^{(n)}(P_{U_0U_1U_2}). \tag{4}$$

It prepares a list with all these pairs of indices $\ell_1 \in [2^{nR_1'}]$ and $\ell_2 \in [2^{nR_2'}]$, and chooses one pair of indices from this list at random. We call the chosen pair (ℓ_1^*, ℓ_2^*). If the list is empty then it chooses (ℓ_1^*, ℓ_2^*) randomly from the set of all indices $[2^{nR_1'}] \times [2^{nR_2'}]$.

The inputs x^n are obtained from the codewords $u_0^n(m_0), u_1^n(m_0, m_1, \ell_1^*), u_2^n(m_0, m_2, \ell_2^*)$ by applying the function f componentwise to these three sequences. That means, for each $j \in [n]$ the j-th channel input is given by

$$x_j = f(u_{0,j}(m_0), u_{1,j}(m_0, m_1, \ell_1^*), u_{2,j}(m_0, m_2, \ell_2^*)$$

where $u_{0,j}(m_0)$, $u_{1,j}(m_0, m_1, \ell_1^*)$, and $u_{2,j}(m_0, m_2, \ell_2^*)$ denote the j-th components of $u_0^n(m_0)$, $u_1^n(m_0, m_1, \ell_1^*)$, and $u_2^n(m_0, m_2, \ell_2^*)$.

3) Decoding: Given that Receiver 1 observes the sequence y_1^n, it forms a list of all the tuples $(\hat{m}_0, \hat{m}_1, \hat{\ell}_1)$ that satisfy

$$(u_0^n(\hat{m}_0), u_1^n(\hat{m}_0, \hat{m}_1, \hat{\ell}_1), y_1^n) \in \mathcal{T}_\epsilon^{(n)}(P_{U_0U_1Y}). \tag{5}$$

It randomly chooses a tuple $(\hat{m}_0, \hat{m}_1, \hat{\ell}_1)$ from this list and produces as its guess $\hat{M}_{c,1} = \hat{m}_0$ and $\hat{M}_{1,p} = \hat{m}_1$. If the list is empty, then it randomly chooses a pair $(\hat{m}_0, \hat{m}_1)$ from $[2^{nR_c}] \times [2^{nR_{1,p}}]$ and guesses $\hat{M}_{c,1} = \hat{m}_0$ and $\hat{M}_{1,p} = \hat{m}_1$.

Receiver 1 finally parses $\hat{M}_{c,1}$ as $(\hat{M}_{0,1}, \hat{M}_{1,c,1}, \hat{M}_{2,c,1})$. Its guess of the messages M_0 and M_1 is then given by $\hat{M}_{0,1}$ and $\hat{M}_1 = (\hat{M}_{1,p}, \hat{M}_{1,c,1})$.

Similarly for Receiver 2.

[2]The choice of $\epsilon/32$ will be helpful later. Here, any $\epsilon' < \epsilon$ suffices.

4) Analysis: We analyze the average probability of error of the above scheme averaged over the random messages, codebooks, and channel realizations. Recall that an error occurs whenever

$$(\hat{M}_{0,1}, \hat{M}_1) \neq (M_0, M_1) \text{ or } (\hat{M}_{0,2}, \hat{M}_2) \neq (M_0, M_2).$$

By the symmetry of the code construction this probability of error equals the average (over all codebooks and channel realizations) probability of error conditioned on the event that $M_c = M_{1,p} = M_{2,p} = 1$, i.e.,

$$\Pr[\text{error}] = \Pr[\text{error}|M_c = M_{1,p} = M_{2,p} = 1].$$

which we analyze in the following. To simplify notation we denote the event that $M_c = M_{1,p} = M_{2,p} = 1$ simply by $\mathcal{M} = 1$.

We define the following events. Let

- $\mathcal{E}_0$ be the event that there is no pair $(\ell_1, \ell_2) \in [2^{nR_1'}] \times [2^{nR_2'}]$ satisfying
$$(U_0^n(1), U_1^n(1, 1, \ell_1), U_2^n(1, 1, \ell_2)) \in \mathcal{T}_{\epsilon/32}^{(n)}(P_{U_0U_1U_2}).$$
- $\mathcal{E}_{0i}$ be the event that
$$(U_0^n(1), U_i^n(1, 1, L_i^*), Y_i^n) \notin \mathcal{T}_\epsilon^{(n)}(P_{U_0U_iY}),$$
where L_1^* and L_2^* denote the pair of indices chosen during the encoding step.
- $\mathcal{E}_{i0}$ be the event that there is a $\hat{m}_0 \neq 1$ such that
$$(U_0^n(\hat{m}_0), U_i^n(\hat{m}_0, 1, L_i^*), Y_i^n) \in \mathcal{T}_\epsilon^{(n)}(P_{U_0U_iY_i}).$$
- $\mathcal{E}_{ii}$ be the event that there is a pair $\hat{m}_i \neq 1$ and $\hat{\ell}_i$ such that
$$(U_0^n(1), U_i^n(1, \hat{m}_i, \hat{\ell}_i), Y_i^n) \in \mathcal{T}_\epsilon^{(n)}(P_{U_0U_iY_i}).$$
- $\mathcal{E}_{i0i}$ be the event that there is a tuple $\hat{m}_0 \neq 1$, $\hat{m}_i \neq 1$, and $\hat{\ell}_i$ such that
$$(U_0^n(\hat{m}_0), U_i^n(\hat{m}_0, \hat{m}_i, \hat{\ell}_i), Y_i^n) \in \mathcal{T}_\epsilon^{(n)}(P_{U_0,U_i,Y}).$$

It is easily verified that

$$\begin{aligned}
&\Pr(\text{error}|\mathcal{M} = 1) \\
&\leq \Pr(\mathcal{E}_0|\mathcal{M} = 1) \\
&\quad + \sum_{i=1}^{2} \Big(\Pr\left(\mathcal{E}_{0i}|\bar{\mathcal{E}}_0, \mathcal{M} = 1\right) + \Pr\left(\mathcal{E}_{i0}|\bar{\mathcal{E}}_{0i}, \mathcal{M} = 1\right) \\
&\qquad + \Pr\left(\mathcal{E}_{ii}|\bar{\mathcal{E}}_{0i}, \mathcal{M} = 1\right) + \Pr\left(\mathcal{E}_{i0i}|\bar{\mathcal{E}}_{0i}, \mathcal{M} = 1\right) \Big).
\end{aligned}$$

We consider each of the terms separately (see also [9], [7]). Hereinafter, a nonnegative function $\delta(\epsilon)$ satisfying $\delta(\epsilon) \to 0$ as $\epsilon \to 0$ can be chosen such that the statements hold.

- By the code construction and by a conditional version of the covering lemma, it follows that
$$\lim_{n \to 0} \Pr(\mathcal{E}_0|\mathcal{M} = 1) = 0, \tag{6}$$
whenever
$$R_1' + R_2' > I(U_1; U_2|U_0) + \delta(\epsilon). \tag{7}$$

- Since the channel outputs Y_i^n is a $P_{Y_i|X}$-i.i.d. sequence given X^n and by the conditional typicality lemma:

$$\lim_{n\to 0} \Pr\left(\mathcal{E}_{0i}|\bar{\mathcal{E}}_0, \mathcal{M}=1\right) = 0. \tag{8}$$

- By the code construction and by the packing lemma:

$$\lim_{n\to 0} \Pr\left(\mathcal{E}_{i0}|\bar{\mathcal{E}}_{0i}, \mathcal{M}=1\right) = 0, \tag{9}$$

whenever

$$R_0 + R_{1,c} + R_{2,c} < I(U_0, U_i; Y_i) - \delta(\epsilon). \tag{10}$$

- By the code construction and by the packing lemma:

$$\lim_{n\to 0} \Pr\left(\mathcal{E}_{ii}|\bar{\mathcal{E}}_{0i}, \mathcal{M}=1\right) = 0, \tag{11}$$

whenever

$$R_{1,p} + R_i' < I(U_i; Y_i|U_0) - \delta(\epsilon). \tag{12}$$

- Again, by the code construction and by the packing lemma:

$$\lim_{n\to 0} \Pr\left(\mathcal{E}_{i0i}|\bar{\mathcal{E}}_{0i}, \mathcal{M}=1\right) = 0, \tag{13}$$

whenever

$$R_0 + R_{1,c} + R_{2,c} + R_{i,p} + R_i' < I(U_0, U_i; Y_i) - \delta(\epsilon). \tag{14}$$

Thus, we conclude that if

$$\begin{aligned} R_1' + R_2' &> I(U_1; U_2|U_0) + \delta(\epsilon) \\ R_0 + R_{1,c} + R_{2,c} &< I(U_0, U_i; Y_i) - \delta(\epsilon) \\ R_{1,p} + R_i' &< I(U_i; Y_i|U_0) - \delta(\epsilon) \\ R_0 + R_{1,c} + R_{2,c} + R_{i,p} + R_i' &< I(U_0, U_i; Y_i) - \delta(\epsilon), \end{aligned} \tag{15}$$

then the probability of error of the scheme tends to 0 as the blocklength n tends to infinity.

It remains to establish the region achieved with the described scheme. Using the Fourier-Motzkin elimination, we obtain that for every rate tuple (R_0, R_1, R_2) satisfying

$$\begin{aligned} R_0 + R_1 &< I(U_0, U_1; Y_1) - \delta(\epsilon) \\ R_0 + R_2 &< I(U_0, U_2; Y_2) - \delta(\epsilon) \\ R_0 + R_1 + R_2 &< I(U_1; Y_1|U_0) + I(U_2; Y_2|U_0) - \delta(\epsilon) \\ &\quad + \min_{i=1,2} I(U_0; Y_i) - I(U_1; U_2|U_0) - \delta(\epsilon) \\ 2R_0 + R_1 + R_2 &< I(U_0, U_1; Y_1) + I(U_0, U_2; Y_2) \\ &\quad - I(U_1; U_2|U_0) - \delta(\epsilon) \end{aligned} \tag{16}$$

for a suitable $\delta(\epsilon) \to 0$ as $\epsilon \to 0$, there exists a choice of the rates $R_0, R_{1,p}, R_{1,c}, R_{2,p}, R_{2,c}, R_1', R_2' > 0$ satisfying (15). The existence of a deterministic coding scheme achieving the same region follows from standard arguments.

It is easy to check that the region above contains the following slightly simpler region[3]:

$$\begin{aligned} R_0 &< \min_i I(U_0; Y_i) - \delta(\epsilon) \\ R_0 + R_1 &< I(U_0, U_1; Y_1) - \delta(\epsilon) \\ R_0 + R_2 &< I(U_0, U_2; Y_2) - \delta(\epsilon) \\ R_0 + R_1 + R_2 &< I(U_1; Y_1|U_0) + I(U_2; Y_2|U_0) \\ &\quad + \min_i I(U_0; Y_i) - I(U_1; U_2|U_0) - \delta(\epsilon) \end{aligned} \tag{17}$$

Thus, all rates (R_0, R_1, R_2) satisfying (17) for some choice of the random variables U_0, U_1, U_2 such that $(U_0, U_1, U_2) \multimap X \multimap (Y_1, Y_2)$ forms a Markov chain are achievable for the BC without feedback. Taking $\epsilon \to 0$, the inclusion $\mathcal{R}_{\text{Marton}} \subseteq \mathcal{C}_{\text{NoFB}}$ is established.

The following two remarks are found useful in the sequel.

Remark 1. *It is easy to check that under conditions* (17)*, there exists an associated choice of scheme parameters such that the associated auxiliary codewords satisfy*

$$\Pr((U_0^n(M_c), U_1^n(M_c, M_{1,p}, L_1^*), U_2^n(M_c, M_{2,p}, L_2^*)) \in \mathcal{T}_{\epsilon/32}^{(n)}(P_{U_0U_1U_2})) \to 1$$

as $n \to \infty$.

Remark 2. *Inspecting the proof, we see that the memoryless channel property has been used only to establish the limit* (8)*. The other limits* (6), (9), (11)*, and* (13) *follow solely from the way we constructed the code. Suppose now we replace the memoryless channel with a general channel* $P_{Y^n|X^n}$*. Then under conditions* (17)*, there exists an associated choice of scheme parameters such that the average error probability goes to zero as* $n \to \infty$*, if*

$$\Pr\left((U_0^n(M_c), U_i^n(M_c, M_{i,p}, L_i^*), Y_i^n) \in \mathcal{T}_\epsilon^{(n)}(P_{U_0U_iY_i})\right) \to 1$$

as $n \to \infty$*, for* $i \in \{1, 2\}$.

V. Lossy Gray-Wyner Coding with Side Information (LGW-SI)

In this section, we introduce a lossy version with side-information of the Gray-Wyner [11] distributed source-coding problem, and derive achievability results. The associated scheme will be used as part of our construction in SectionVI. The lossless counterpart of this problem was studied in [12].

A. Setup and Achievable Regions

Our setup is parameterized by the tuple $(\mathcal{X}, \mathcal{Y}_1, \mathcal{Y}_2, \mathcal{V}_1, \mathcal{V}_2, P_{XY_1Y_2}, P_{V_1|X}, P_{V_2|X}, n)$, where

- $\mathcal{X}, \mathcal{Y}_1, \mathcal{Y}_2, \mathcal{V}_1, \mathcal{V}_2$ are discrete finite alphabets;
- $P_{XY_1Y_2}$ is a joint probability distribution over the alphabet $\mathcal{X} \times \mathcal{Y}_1 \times \mathcal{Y}_2$;
- $P_{V_1|X}$ and $P_{V_2|X}$ are conditional probability distributions over $\mathcal{V}_1$ and $\mathcal{V}_2$ given some random variable $X \in \mathcal{X}$;
- n is the blocklength.

[3]In fact, it was shown [10] that the associated unions (over all permissible choices of auxiliary random variables) coincide.

In the following let $\{(X_t, Y_{1,t}, Y_{2,t})\}_{t=1}^n$ be an i.i.d. sequence of triplets of discrete random variables, with marginal distribution $P_{XY_1Y_2}$. Consider a distributed source coding setting where a sender observes the sequence X^n, Receiver 1 observes the side-information Y_1^n, and Receiver 2 observes the side-information Y_2^n. It is assumed that the sender can noiselessly send three rate-limited messages K_0, K_1, K_2 to the receivers: a common message K_0 to both receivers, a private message K_1 to Receiver 1 only, and another private message K_2 to Receiver 2 only. More precisely, the encoding procedure is described by an encoding function $\lambda^{(n)} \colon \mathcal{X}^n \to [2^{nR_0}] \times [2^{nR_1}] \times [2^{nR_2}]$, which for a sequence X^n produces the messages $(K_0, K_1, K_2) = \lambda^{(n)}(X^n)$. Each Receiver i, for $i \in \{1, 2\}$, produces a reconstruction sequence $\hat{V}_i^n$ based on the observed messages K_0 and K_i and the side-information Y_i^n. To this end, it applies a reconstruction function $\Lambda_i^{(n)} \colon [2^{nR_0}] \times [2^{nR_i}] \times \mathcal{Y}_i^n \to \mathcal{V}_i^n$ to the messages and the side-information, i.e., it produces $\hat{V}_i^n = \Lambda_i^{(n)}(K_0, K_i, Y_i^n)$. The goal of the communication is that for each $i \in \{1, 2\}$, the reconstruction sequence $\hat{V}_i^n$ is jointly typical with X^n according to $P_X \times P_{V_i|X}$.

A rate triplet (R_0, R_1, R_2) is said to be *ϵ-achievable* if there exist a sequence of encoding and reconstruction functions $(\lambda^{(n)}, \Lambda_1^{(n)}, \Lambda_2^{(n)})$ such that:

$$\Pr\left((X^n, \hat{V}_i^n) \notin \mathcal{T}_\epsilon^n(P_{XV_i})\right) \to 0$$

as $n \to \infty$, for $i \in \{1, 2\}$. A triplet is said to be *achievable* if it is ϵ-achievable for all $\epsilon > 0$. The closure of the set of all achievable rate triplets is denoted $\mathcal{R}_{\text{LGW}}$.

Let $\mathcal{R}_{\text{LGW}}^{\text{inner}}$ be the closure of the set of all nonnegative rate triplets (R_0, R_1, R_2) satisfying

$$R_1 > I(X; V_1|V_0, Y_1) \tag{18a}$$
$$R_2 > I(X; V_2|V_0, Y_2) \tag{18b}$$
$$R_0 + R_1 > I(X; V_0, V_1|Y_1) \tag{18c}$$
$$R_0 + R_2 > I(X; V_0, V_2|Y_2), \tag{18d}$$

for some choice of the random variable V_0 such that

$$(V_0, V_1, V_2) \multimap X \multimap (Y_1, Y_2). \tag{19}$$

Let $\mathcal{R}_{\text{LGW}}^{\text{inner*}}$ be the closure of the set of all nonnegative rate triplets (R_0, R_1, R_2) satisfying

$$R_0 > \max_i I(X; V_0|Y_i), \tag{20a}$$
$$R_1 > I(X; V_1|V_0, Y_1) \tag{20b}$$
$$R_2 > I(X; V_2|V_0, Y_2) \tag{20c}$$

for some choice of V_0 satisfying (19).

Theorem 2. $\mathcal{R}_{\text{LGW}}^{\text{inner*}} \subseteq \mathcal{R}_{\text{LGW}}^{\text{inner}} \subseteq \mathcal{R}_{\text{LGW}}$. *Furthermore,* $\mathcal{R}_{\text{LGW}}^{\text{inner}}$ *is convex.*

Proof. Inclusion $\mathcal{R}_{\text{LGW}}^{\text{inner}} \subseteq \mathcal{R}_{\text{LGW}}$ is established by the scheme and the analysis in Section V-B. Inclusion $\mathcal{R}_{\text{LGW}}^{\text{inner*}} \subseteq \mathcal{R}_{\text{LGW}}^{\text{inner}}$ follows by noting that for a given choice of V_0, the region defined through constraints (20) is contained in the region defined through (18). This holds because Constraints (20b) and (20c) coincide with Constraints (18a) and (18b), the two constraints (20a) and (20b) imply the single constraint (18c), and the two constraints (20a) and (20c) imply constraint (18d).

The convexity of $\mathcal{R}_{\text{LGW}}^{\text{inner}}$ is proved in Appendix B. ■

Remark 3. *The possibly smaller region $\mathcal{R}_{\text{LGW}}^{\text{inner*}}$ is given here for reasons of convenience, as it allows us to later simplify our achievable region for the DMBC with feedback introduced in Section VI, making it easier to evaluate.*

B. Scheme

In this section we describe a scheme achieving $\mathcal{R}_{\text{LGW}}^{\text{inner}}$ for the LGW-SI problem with parameters $\mathcal{X}$, $\mathcal{Y}_1$, $\mathcal{Y}_2$, $\mathcal{V}_1$, $\mathcal{V}_2$, $P_{XY_1Y_2}$, $P_{V_1|X}$, $P_{V_2|X}$. Our scheme has parameters $(\mathcal{V}_0, P_{V_0V_1V_2|X}, R_0, R_1, R_2, R_0', R_1', R_2', \epsilon)$, where

- $\mathcal{V}_0$ is an auxiliary alphabet;
- $P_{V_0V_1V_2|X}$ is a conditional joint probability distribution over $\mathcal{V}_0 \times \mathcal{V}_1 \times \mathcal{V}_2$ given some $X \in \mathcal{X}$ such that its marginals satisfy $\sum_{v_0,v_2} P_{V_0V_1V_2|X}(v_0, v_1, v_2|x) = P_{V_1|X}(v_1|x)$ and $\sum_{v_0,v_1} P_{V_0V_1V_2|X}(v_0, v_1, v_2|x) = P_{V_2|X}(v_2|x)$;
- $R_0, R_1, R_2, R_0', R_1', R_2' > 0$ are nonnegative rates;
- $\epsilon > 0$ is a small number; and
- n is the scheme's blocklength.

1) Codebook Generation: Independently generate three codebooks $\mathcal{C}_0, \mathcal{C}_1, \mathcal{C}_2$. Each codebook $\mathcal{C}_i$, for $i \in \{0, 1, 2\}$, consists of $\lfloor 2^{nR_i} \rfloor$ bins with $\lfloor 2^{nR_i'} \rfloor$ codewords of length n. All the entries of all the codewords in codebook $\mathcal{C}_i$ are randomly and independently drawn according to P_{V_i}. We denote the codewords in bin $m_i \in [2^{nR_i}]$ by $\{v_i^n(m_i, \ell_i)\}_{\ell_i=1}^{\lfloor 2^{nR_i'} \rfloor}$.

All codebooks are revealed to the sender; Codebooks $\{\mathcal{C}_0, \mathcal{C}_i\}$ are revealed to Receiver i, for $i \in \{1, 2\}$.

2) LGW-SI Encoder: Given that the sender observes the sequence $X^n = x^n$, it searches the codebooks $\mathcal{C}_0, \mathcal{C}_1, \mathcal{C}_2$ for a triplet of codewords $v_0^n(k_0, \ell_0) \in \mathcal{C}_0$, $v_1^n(k_1, \ell_1) \in \mathcal{C}_1$, $v_2^n(k_2, \ell_2) \in \mathcal{C}_2$ such that for $i \in \{1, 2\}$:[4]

$$(X^n, v_0^n(k_0, \ell_0), v_i^n(k_1, \ell_1)) \in \mathcal{T}_{\epsilon/2}^n(P_{XV_0V_i}). \tag{21}$$

It then forms a list of all tuples of indices $(k_0, \ell_0, k_1, \ell_1, k_2, \ell_2)$ satisfying (21). If the list is non-empty, the sender randomly chooses one tuple from this list at random. If the list is empty, it randomly chooses a tuple $(k_0, \ell_0, k_1, \ell_1, k_2, \ell_2)$ from the set $[2^{nR_0}] \times [2^{nR_0'}] \times [2^{nR_1}] \times [2^{nR_1'}] \times [2^{nR_2}] \times [2^{nR_2'}]$. We denote the chosen indices by $k_0^*, \ell_0^*, k_1^*, \ell_1^*, k_2^*, \ell_2^*$.

The sender sends the index k_0^* over the common link, and the indices k_1^*, k_2^* over the respective private links.

3) LGW-SI Decoder: Given that Receiver i observes the side-information $Y_i^n = y_i^n$, the common message $K_0 = k_0$, and the private message $K_i = k_i$, it seeks a codeword $v_0^n(k_0, \ell_0)$ in the k_0-th bin of codebook $\mathcal{C}_0$ and a codeword $v_i^n(k_i, \ell_i)$ in the k_i-th bin of codebook $\mathcal{C}_i$ such that $(v_0^n(k_0, \ell_0), v_i^n(k_i, \ell_i), y_i^n) \in \mathcal{T}_\epsilon^n(P_{V_0V_iY_i})$. If exactly one such pair of codewords exists, Receiver i produces as its reconstruction sequence $\hat{V}_i^n = v_i^n(k_i, \ell_i)$. Otherwise, it randomly chooses a pair (k_i', ℓ_i') from the set $[2^{nR_i}] \times [2^{nR_i'}]$ and produces the reconstruction sequence $\hat{V}_i^n = v_i^n(k_i', \ell_i')$.

[4]One could employ a more elaborate construction (based on the multivariate covering Lemma [9]) where the codeword triplet is chosen jointly. However, this alternative scheme does not lead to a larger achievable region.

4) Analysis: We analyze the failure probability $\Pr\left(\mathcal{E}^{(1)} \cup \mathcal{E}^{(2)}\right)$ associated with the above random coding scheme, where $\mathcal{E}^{(i)}$ is the event where Receiver i fails, i.e., where $(X^n, \hat{V}_i^n) \notin \mathcal{T}_\epsilon^n(P_{XV_i})$.

In what follows, let $K_0^*, L_0^*, K_1^*, L_1^*, K_2^*, L_2^*$ be the tuple of indices chosen by the sender. Also, let

- $\mathcal{E}_0$ be the event that $X^n \notin T_{\epsilon/8}^n(P_X)$;
- $\mathcal{E}_1$ be the event that
$$\forall k_0, \ell_0 : \ (X^n, V_0^n(k_0, \ell_0)) \notin T_{\epsilon/4}^n(P_{XV_0});$$
- $\mathcal{E}_{2,i}$, for $i \in \{1, 2\}$, be the event that
$$\forall k_i, \ell_i : \ (X^n, V_0^n(K_0^*, L_0^*), V_i^n(k_i, \ell_i)) \notin T_{\epsilon/2}^n(P_{XV_0V_i});$$
- $\mathcal{E}_{3,i}$, for $i \in \{1, 2\}$, be the event that
$$(V_0^n(K_0^*, L_0^*), V_i^n(K_i^*, L_i^*), Y_i^n) \notin T_\epsilon^n(P_{V_0V_iY_i});$$
- $\mathcal{E}_{4,i}$, for $i \in \{1, 2\}$, be the event that
$$\exists \ell_i \neq 1 : \quad (V_0^n(K_0^*, L_0^*), V_i^n(K_i^*, \ell_i), Y_i^n) \in T_\epsilon^n(P_{V_0V_iY_i});$$
- $\mathcal{E}_{5,i}$, for $i \in \{1, 2\}$, be the event that
$$\exists \ell_0, \ell_i \neq 1 : \quad (V_0^n(K_0^*, \ell_0), V_i^n(K_i^*, \ell_i), Y_i^n) \in T_\epsilon^n(P_{V_0V_iY_i}).$$

Notice that whenever event $(\mathcal{E}_0^c \cap \mathcal{E}_1^c \cap \mathcal{E}_{2,i}^c)$ occurs, then $(X^n, V_i^n(K_i^*, L_i^*)) \in T_\epsilon^n(P_{XV_i})$. If also event $(\mathcal{E}_{3,i}^c \cap \mathcal{E}_{4,i}^c \cap \mathcal{E}_{5,i}^c)$ occurs, then Receiver i produces $\hat{V}_i^n = V_i^n(K_i^*, L_i^*)$. Therefore, denoting by $\mathcal{E}^{(i)}$ the event where Receiver i fails, we have:

$$\begin{aligned}\Pr(\mathcal{E}^{(i)}) &\leq \Pr\left(\mathcal{E}_0 \cup \mathcal{E}_1 \cup \mathcal{E}_{2,i} \cup \mathcal{E}_{3,i} \cup \mathcal{E}_{4,i} \cup \mathcal{E}_{5,i}\right) \\ &\leq \Pr(\mathcal{E}_0) + \Pr(\mathcal{E}_1|\mathcal{E}_0^c) \\ &\quad + \Pr(\mathcal{E}_{2,i}|\mathcal{E}_1^c) + \Pr(\mathcal{E}_{3,i}|\mathcal{E}_{2,i}^c) \\ &\quad + \Pr(\mathcal{E}_{4,i}) + \Pr(\mathcal{E}_{5,i}). \end{aligned} \tag{22}$$

We analyze each of the summands separately. Hereinafter, a nonnegative function $\delta(\epsilon)$ satisfying $\delta(\epsilon) \to 0$ as $\epsilon \to 0$, can be chosen such that the statements hold.

- Since X^n is P_X-i.i.d. and by the weak law of large numbers:
$$\lim_{n\to\infty} \Pr\left(\mathcal{E}_0\right) = 0 \tag{23}$$
- By the code construction and the covering lemma [9]:
$$\lim_{n\to\infty} \Pr\left(\mathcal{E}_1|\mathcal{E}_0^c\right) = 0 \tag{24}$$
whenever
$$R_0' + R_0 > I(X; V_0) + \delta(\epsilon) \tag{25}$$
- Again, by the code construction and the covering lemma:
$$\lim_{n\to\infty} \Pr\left(\mathcal{E}_{2,i}|\mathcal{E}_1^c\right) = 0 \tag{26}$$
whenever
$$R_i' + R_i > I(V_i; X, V_0) + \delta(\epsilon) \tag{27}$$
- The pair $V_0^n(K_0^*, L_0^*), V_i^n(K_i^*, L_i^*)$ depends on Y_i^n only through X^n, i.e., the Markov chain
$$(V_0^n(K_0^*, L_0^*), V_i^n(K_i^*, L_i^*)) \multimap X^n \multimap Y_i^n$$
holds. Therefore, because Y_i^n is $P_{Y_i|X}$-independent given X^n, by the conditional typicality lemma:
$$\lim_{n\to\infty} \Pr\left(\mathcal{E}_{3,i}|\mathcal{E}_{2,i}^c\right) = 0. \tag{28}$$
- Notice that the codewords $\{V_i^n(K_i^*, \ell_i)\}$ for $\ell_i \in [2^{nR_i'}] \backslash \{L_i^*\}$ are not independent and P_{V_i}-i.i.d.[5] In Appendix A we prove Inequality (29) on top of the next page, where on the right-hand side we have the probability that one of the $\lfloor 2^{nR_i'} \rfloor$ independent and P_{V_i}-i.i.d. codewords $\{V_i^n(1, \ell_i)\}_{\ell_i=1}^{\lfloor 2^{nR_i'} \rfloor}$ is jointly ϵ-typical with the pair $(V_0^n(K_0^*, L_0^*), Y_i^n)$. By the packing lemma this probability tends to 0 as n tends to ∞ whenever
$$R_i' < I(V_i; V_0Y_i) - \delta(\epsilon) \tag{31}$$
We thus conclude that
$$\lim_{n\to\infty} \Pr\left(\mathcal{E}_{4,i}\right) = 0, \tag{32}$$
whenever (31) holds.
- Following similar steps as in Appendix A, upper bound (30) can be proved. Then, by the multivariate packing lemma (Lemma 4):
$$\lim_{n\to\infty} \Pr\left(\mathcal{E}_{5,i}\right) = 0, \tag{33}$$
whenever
$$R_0' + R_i' < I(V_0; Y_i) + I(V_i; V_0Y_i) - \delta(\epsilon) \tag{34}$$

Combining (22) with (25), (27), (31), and (34) we obtain that $\Pr\left(\mathcal{E}^{(i)}\right) \to 0$ as $n \to \infty$ whenever:

$$\begin{aligned} R_0' + R_0 &> I(X; V_0) + \delta(\epsilon) \\ R_1' + R_1 &> I(V_1; X, V_0) + \delta(\epsilon) \\ R_2' + R_2 &> I(V_2; X, V_0) + \delta(\epsilon) \\ R_0' + R_1' &< I(V_0; Y_1) + I(V_1; V_0, Y_1) - \delta(\epsilon) \\ R_0' + R_2' &< I(V_0; Y_2) + I(V_2; V_0, Y_2) - \delta(\epsilon) \\ R_0' &< \min\{I(V_0; Y_1), I(V_0; Y_2)\} - \delta(\epsilon) \\ R_1' &< I(V_1; V_0, Y_1) - \delta(\epsilon) \\ R_2' &< I(V_2; V_0, Y_2) - \delta(\epsilon) \end{aligned} \tag{35}$$

We now argue that with an appropriate choice of the parameters our scheme achieves the region $\mathcal{R}_{\text{LGW}}^{\text{inner}}$. Employing the Fourier-Motzkin elimination algorithm to eliminate the nuisance variables (R_0', R_1', R_2'), we obtain that if (R_0, R_1, R_2) satisfies

$$\begin{aligned} R_1 &> I(V_1; X, V_0) - I(V_1; V_0, Y_1) + \delta(\epsilon) \\ R_2 &> I(V_2; X, V_0) - I(V_2; V_0, Y_2) + \delta(\epsilon) \\ R_0 + R_1 &> I(X; V_0) + I(V_1; X, V_0) - I(V_0; Y_1) \\ &\quad - I(V_1; V_0, Y_1) + \delta(\epsilon) \\ R_0 + R_2 &> I(X; V_0) + I(V_2; X, V_0) - I(V_0; Y_2) \\ &\quad - I(V_2; V_0, Y_2) + \delta(\epsilon) \end{aligned}$$

[5]This can be seen with the following simple example. Let the heights of two students A_0 and A_1 be uniformly distributed over the interval $[1.7, 1.9]$ m and independent of each other. Also, let C be the index of the student that has height larger than 1.89m if this index is unique; otherwise let C be Bern($\frac{1}{2}$). Let $\bar{C}$ be the index in $\{0, 1\}$ not equal to C. Notice that $\Pr\left(A_0 \geq 1.89\right) = \frac{1}{20}$, whereas $\Pr\left(A_{\bar{C}} \geq 1.89\right) = \Pr\left(A_0 \geq 1.89 \text{ and } A_1 \geq 1.89\right) = \frac{1}{400}$. Thus, $A_{\bar{C}}$ is not uniform over $[1.7, 1.9]$.

$$\Pr\left(\mathcal{E}_{4,i}\right) \leq \Pr\left(\bigcup_{\ell_i=1}^{\lfloor 2^{nR_i'}\rfloor} (V_0^n(K_0^*, L_0^*), V_i^n(1,\ell_i), Y_i^n) \in T_\epsilon^n(P_{V_0V_iY_i})\right) \tag{29}$$

$$\Pr\left(\mathcal{E}_{5,i}\right) \leq \Pr\left(\bigcup_{\substack{\ell_0\in[2^{nR_0'}],\ \ell_i\in[2^{nR_i'}]\\ (\ell_0,\ell_1)\neq(1,1)}} (V_0^n(1,\ell_0), V_i^n(1,\ell_i), Y_i^n) \in T_\epsilon^n(P_{V_0V_iY_i})\right) \tag{30}$$

or equivalently (due of the Markov chain (V_0,V_1,V_2)—∘—X—∘—(Y_1,Y_2)), if

$$\begin{aligned} R_1 &> I(X;V_1|V_0,Y_1)+\delta(\epsilon)\\ R_2 &> I(X;V_2|V_0,Y_2)+\delta(\epsilon)\\ R_0+R_1 &> I(X;V_0|Y_1)+I(X;V_1|V_0,Y_1)+\delta(\epsilon)\\ R_0+R_2 &> I(X;V_0|Y_2)+I(X;V_2|V_0,Y_2)+\delta(\epsilon), \end{aligned} \tag{36}$$

then there exists a choice of (R_0', R_1', R_2') such that the tuple $(R_0,R_1,R_2,R_0',R_1',R_2')$ satisfies (35). Thus, we conclude that the region (36) is ϵ-achievable for all choices of the auxiliary random variable V_0 satisfying the Markov chain (V_0,V_1,V_2)—∘—X—∘—(Y_1,Y_2). Since (36) coincides with (18) as $\epsilon \to 0$, the achievability of $\mathcal{R}_{\text{LGW}}^{\text{inner}}$ is established. The existence of a deterministic coding scheme achieving the same region follows from standard arguments.

The following remark is found useful in the sequel.

Remark 4. *In our error analysis, only Limits* (23) *and* (28) *rely on the assumption that* (X^n, Y_1^n, Y_2^n) *are* $P_{XY_1Y_2}$*-i.i.d. It is easy to check that replacing this assumption with the more general assumptions*

(i) $\Pr(X^n \in \mathcal{T}_{\epsilon/8}^n(P_X)) \to 1$ *as* $n\to\infty$.

(ii) (Y_1^n, Y_2^n) *is* $P_{Y_1Y_2|X}$*-independent given* X^n.

still guarantees the existence of associated parameters such that the scheme above ϵ*-achieves the region (36). In particular,*

$$\Pr\left((X^n, V_i^n(K_i^*, L_i^*)) \notin T_\epsilon^n(P_{XV_i})\right) \to\ 0$$

and

$$\Pr\left(\hat{V}_i^n \neq V_i^n(K_i^*, L_i^*)\right) \to 0,$$

for $i\in\{1,2\}$*, as* $n\to\infty$.

VI. Main Result

A. Achievable Region

Consider a DMBC with generalized feedback, given by $P_{Y_1,Y_2,\tilde{Y}|X}$. Let $\mathcal{R}_{\text{inner}}$ be the closed convex hull of the set of all nonnegative triplets (R_0,R_1,R_2) that satisfy Inequalities (37) shown on top of the next page, for some choice of auxiliary random variables $(U_0,U_1,U_2,V_0,V_1,V_2)$ and function f such that

$$(U_0,U_1,U_2)\text{—∘—}X\text{—∘—}(Y_1,Y_2,\widetilde{Y})$$
$$(V_0,V_1,V_2)\text{—∘—}(U_0,U_1,U_2,\widetilde{Y})\text{—∘—}(Y_1,Y_2)$$

where $X = f(U_0,U_1,U_2)$ and $(Y_1,Y_2,\tilde{Y}) \sim P_{Y_1,Y_2,\tilde{Y}|X}$.

Let $\mathcal{R}_{\text{inner}}^*$ be the closed convex hull of the set of all nonnegative triplets (R_0,R_1,R_2) that satisfy Inequalities (38), for some choice of auxiliary random variables $(U_0,U_1,U_2,V_0,V_1,V_2)$ and function f such that

$$(U_0,U_1,U_2)\text{—∘—}X\text{—∘—}(Y_1,Y_2,\widetilde{Y})$$
$$(V_0,V_1,V_2)\text{—∘—}(X,\widetilde{Y})\text{—∘—}(Y_1,Y_2,U_0,U_1,U_2)$$

where $X = \text{f}(U_0,U_1,U_2)$ and $(Y_1,Y_2,\tilde{Y}) \sim P_{Y_1,Y_2,\tilde{Y}|X}$.

Theorem 3. $\mathcal{R}_{\text{inner}}^* \subseteq \mathcal{R}_{\text{inner}} \subseteq \mathcal{C}_{\text{GenFB}}$.

The proof of the Theorem is given in Subsection VI-B. A few remarks are in order:

Remark 5. *The possibly smaller region* $\mathcal{R}_{\text{inner}}^*$ *is given here for the sake of the examples in Section VII, being easier to evaluate.*

Remark 6. *Both* $\mathcal{R}_{\text{inner}}$ *and* $\mathcal{R}_{\text{inner}}^*$ *include* $\mathcal{R}_{\text{Marton}}$*, because when for a given choice of* (U_0,U_1,U_2)*, constraints* (37) *or* (38) *are specialized to* $(V_0,V_1,V_2)=\text{const}$*, then they result in the Marton-region* (3)*. The inclusion is also clear from the construction of our scheme in Subsection VI-B ahead.*

Remark 7. *Recall that for fixed finite alphabets, the Shannon information measures are continuous (say w.r.t. Euclidean distance) in the joint distribution [13]. Fix the channel's input, output and feedback alphabets. Then for any fixed choice of* $(P_{U_0U_1U_2}, f, P_{V_0V_1V_2|U_0U_1U_2\widetilde{Y}})$*, the quantities on the right-hand-side of the Inequalities (37) are continuous in* $P_{Y_1,Y_2,\widetilde{Y}|X}$*. A similar statement holds for the Inequalities (38).*

Remark 8. *Following the previous remark, we conclude that for any DMBC* $P_{Y_1Y_2|X}$ *with feedback alphabet* $\widetilde{\mathcal{Y}} = \mathcal{Y}_1\times\mathcal{Y}_2$*, if the region* $\mathcal{R}_{\text{inner}}$ *(resp.* $\mathcal{R}_{\text{inner}}^*$*) associated with noiseless feedback strictly contains* $\mathcal{C}_{\text{NoFB}}$*, then* $\mathcal{R}_{\text{inner}}$ *(resp.* $\mathcal{R}_{\text{inner}}^*$*) associated with noisy/generalized feedback strictly contains* $\mathcal{C}_{\text{NoFB}}$ *as well for or any choice of the feedback channel* $P_{\widetilde{Y}|XY_1Y_2}$ *that is close enough (say in Euclidean distance) to the noiseless case (i.e., to* $\widetilde{Y}=(Y_1,Y_2)$*),*

B. Scheme achieving $\mathcal{R}_{\text{innner}}$

Our scheme combines Marton's no-feedback scheme of Section IV-B with our LGW-SI scheme of Section V-B using a block-Markov framework. We first present the high-level idea of the scheme, which is also depicted in Figure 1. Transmission takes place over $B+1$ consecutive blocks, where the first B blocks are of length n each, and the

$$
\begin{aligned}
R_0 + R_1 &\leq I(U_0, U_1; Y_1, V_1) - I(U_0, U_1, U_2, \widetilde{Y}; V_0, V_1|Y_1) \\
R_0 + R_2 &\leq I(U_0, U_2; Y_2, V_2) - I(U_0, U_1, U_2, \widetilde{Y}; V_0, V_2|Y_2) \\
R_0 + R_1 + R_2 &\leq I(U_1; Y_1, V_1|U_0) + I(U_2; Y_2, V_2|U_0) + \min_{i\in\{1,2\}} I(U_0; Y_i, V_i) - I(U_1; U_2|U_0) \\
&\quad - I(U_0, U_1, U_2, \widetilde{Y}; V_1|V_0, Y_1) - I(U_0, U_1, U_2, \widetilde{Y}; V_2|V_0, Y_2) - \max_{i\in\{1,2\}} I(U_0, U_1, U_2, \widetilde{Y}; V_0|Y_i) \\
2R_0 + R_1 + R_2 &\leq I(U_1; Y_1, V_1|U_0) + I(U_2; Y_2, V_2|U_0) + 2\min_{i\in\{1,2\}} I(U_0; Y_i, V_i) - I(U_1; U_2|U_0) \\
&\quad - I(U_0, U_1, U_2, \widetilde{Y}; V_1|V_0, Y_1) - I(U_0, U_1, U_2, \widetilde{Y}; V_2|V_0, Y_2)
\end{aligned} \tag{37}
$$

$$
\begin{aligned}
R_0 &\leq \min_{i\in\{1,2\}} I(U_0; Y_i, V_i) - \max_{i\in\{1,2\}} I(V_0; X, \widetilde{Y}|Y_i) \\
R_0 + R_1 &\leq I(U_0, U_1; Y_1, V_1) - I(X, \widetilde{Y}; V_1|V_0, Y_1) - \max_{i\in\{1,2\}} I(V_0; X, \widetilde{Y}|Y_i) \\
R_0 + R_2 &\leq I(U_0, U_2; Y_2, V_2) - I(X, \widetilde{Y}; V_2|V_0, Y_2) - \max_{i\in\{1,2\}} I(V_0; X, \widetilde{Y}|Y_i) \\
R_0 + R_1 + R_2 &\leq I(U_1; Y_1, V_1|U_0) + I(U_2; Y_2, V_2|U_0) + \min_{i\in\{1,2\}} I(U_0; Y_i, V_i) \\
&\quad - I(U_1; U_2|U_0) - I(X, \widetilde{Y}; V_1|V_0, Y_1) - I(X, \widetilde{Y}; V_2|V_0, Y_2) - \max_{i\in\{1,2\}} I(V_0; X, \widetilde{Y}|Y_i)
\end{aligned} \tag{38}
$$

Fig. 1. Block-Markov strategy of our feedback-scheme.

last block is of length $n' = \gamma n$ for $\gamma > 1$. We denote the input/output/feedback sequences in Block $b \in [B]$ by $X^n_{(b)}, Y^n_{i,(b)}, \widetilde{Y}^n_{(b)}$, respectively, and the input/output sequences in Block $B+1$ by $X^{n'}_{(B+1)}, Y^{n'}_{i,(B+1)}$. The messages to be sent are in a product form $M_i = (M_{i,(1)}, \ldots, M_{i,(B)})$, for $i \in \{0, 1, 2\}$, where each $M_{i,(b)}$ is uniformly distributed over the set $[2^{nR_i}]$.

In each block b the transmitter uses a no-feedback Marton scheme to send the Messages $M_{0,(b)}, M_{1,(b)}, M_{2,(b)}$ together with update information $K_{0,(b-1)}, K_{1,(b-1)}, K_{2,(b-1)}$ pertaining to the messages sent in the previous block. An exception is the first (resp. last) block where only the message tuple (resp. update information) is sent. Loosely speaking, the update information is constructed in a way that when $(K_{0,(b)}, K_{i,(b)})$ is available at Receiver i, the latter can use it to "improve" its observations $Y^n_{i,(b)}$ in that block. This facilitates the decoding of the corresponding messages $M_{0,(b)}, M_{1,(b)}, M_{2,(b)}$, which otherwise might not have been possible to decode reliably. The update information is generated via an LGW-SI code when the encoder observes the auxiliary Marton-codewords and the feedback signal and the receivers' side-information consist of its channel outputs.

Each Receiver i, for $i \in \{1, 2\}$, performs backward decoding. It starts from the last block and decodes the update information $(K_{0,(B)}, K_{i,(B)})$ based on $Y^{n'}_{i,(B+1)}$. Denote its guess by $\hat{K}_{0,i,(B)}, \hat{K}_{i,(B)}$. Then, for each block $b \in [B]$, starting from block B and going backwards, it performs the following steps:

1) Using $(\hat{K}_{0,b}, \hat{K}_{i,b})$, it "improves" its block-b outputs $Y^n_{i,(b)}$.
2) Based on these "improved" outputs, it then decodes the data $(M_{0,(b)}, M_{i,(b)})$ and the update information $(K_{0,(b-1)}, K_{i,(b-1)})$. We denote the corresponding guesses by $(\hat{M}_{0,(b)}, \hat{M}_{i,(b)})$ and $(\hat{K}_{0,i,(b-1)}, \hat{K}_{i,(b-1)})$.

We now describe the coding scheme in more detail. Our scheme has parameters ($\mathcal{U}_0$, $\mathcal{U}_1$, $\mathcal{U}_2$,$\mathcal{V}_0$,$\mathcal{V}_1$,$\mathcal{V}_2$, P_{U_0,U_1,U_2}, $P_{V_0V_1V_2|U_0,U_1,U_2,\widetilde{Y}}$, f, R_0, R_1, R_2, $\bar{R}'_1$, $\bar{R}'_2$,$\tilde{R}_0$, $\tilde{R}_1$, $\tilde{R}_2$, $\tilde{R}'_0$, $\tilde{R}'_1$, $\tilde{R}'_2$, ϵ, γ, n, B), where:

- $\mathcal{U}_0, \mathcal{U}_1, \mathcal{U}_2, \mathcal{V}_0, \mathcal{V}_1$, and $\mathcal{V}_2$ are finite auxiliary alphabets;
- P_{U_0,U_1,U_2} is a joint probability law over $\mathcal{U}_0 \times \mathcal{U}_1 \times \mathcal{U}_2$;
- $P_{V_0V_1V_2|U_0,U_1,U_2,\widetilde{Y}}$ is a conditional probability law over $\mathcal{V}_0 \times \mathcal{V}_1 \times \mathcal{V}_2$ given a tuple $(U_0, U_1, U_2, \widetilde{Y})$;
- f is a function $f : \mathcal{U}_0 \times \mathcal{U}_1 \times \mathcal{U}_2 \to \mathcal{X}$;
- $R_0, R_1, R_2, \tilde{R}_0, \tilde{R}_1, \tilde{R}_2$ are nonnegative communication rates;
- $\bar{R}'_1, \bar{R}'_2, \tilde{R}'_0, \tilde{R}'_1, \tilde{R}'_2$ are nonnegative binning rates;
- $\epsilon > 0$ is a small number; and
- n, γ, and B are large positive integers determining the scheme's blocklength.

Our scheme is of rates R_0, R_1, R_2 and of blocklength $N = Bn + \gamma n$. Before the transmission starts, we divide the blocklength N into $B+1$ blocks: the first B blocks are of length n and the last block is of length $n' \stackrel{\text{def}}{=} \gamma n$. We also construct the following codes.

For each block $b \in [B]$ we construct a Marton code $\left(\mathcal{C}_{0,(b)}^{(\text{Marton})}, \{\mathcal{C}_{1,(b)}^{(\text{Marton})}(m_{0,b})\}, \{\mathcal{C}_{2,(b)}^{(\text{Marton})}(m_{0,b})\}\right)$ for a DMBC with parameters $(\mathcal{X}, \mathcal{Y}_1 \times \mathcal{V}_1, \mathcal{Y}_2 \times \mathcal{V}_2, P_{Y_1Y_2V_1V_2|X})$ using the code construction in Subsection IV-B1. As parameters of this construction we choose:

- the auxiliary alphabets $\mathcal{U}_0, \mathcal{U}_1, \mathcal{U}_2$;
- the joint law P_{U_0,U_1,U_2} over these alphabets;
- the function $f\colon \mathcal{U}_0 \times \mathcal{U}_1 \times \mathcal{U}_2 \to \mathcal{X}$;
- the nonnegative communication rates $\bar{R}_0 = R_0 + \tilde{R}_0, \bar{R}_1 = R_1 + \tilde{R}_1, \bar{R}_2 = R_2 + \tilde{R}_2$, where $\bar{R}_1, \bar{R}_1$ are split into a common and a private part in a suitable fashion;
- the nonnegative binning rates $\bar{R}_1', \bar{R}_2'$;
- the small number ϵ; and
- the blocklength n.

For block $B+1$, we use an arbitrary scheme (e.g., a suitable Marton scheme) of block length n' for the DMBC with parameters $(\mathcal{X}, \mathcal{Y}_1, \mathcal{Y}_2, P_{Y_1Y_2|X})$, that achieves the rate triplet $(\gamma^{-1}\widetilde{R}_0, \gamma^{-1}\widetilde{R}_1, \gamma^{-1}\widetilde{R}_2)$ with a vanishing probability of error as $n' \to \infty$. To allow this, we assume throughout the proof that the single-user capacities C_1 and C_2 of $P_{Y_1|X}$ and $P_{Y_2|X}$ are both positive[6]. Under this assumption, it is readily verified that for $\gamma > 1$ large enough such a scheme exists.

In what follows, let $\varphi_{(b)}, \Phi_{1,(b)}, \Phi_{2,(b)}$ denote the encoding and decoding rules corresponding to the code in block b, for any $b \in [B+1]$. Also, let the triplet $(U_{0,(b)}^n, U_{1,(b)}^n, U_{2,(b)}^n)$ denote the auxiliary codewords produced by the block-b Marton encoder $\varphi_{(b)}$, for any $b \in [B]$. Accordingly, let $X_{(b)}^n, Y_{1,(b)}^n, Y_{2,(b)}^n$ and $\widetilde{Y}_{(b)}^n$ denote the corresponding channel input, channel outputs, and feedback output, respectively.

For blocks $b \in [B]$ we also construct a LWG-SI code $\left(\mathcal{C}_{0,(b)}^{(\text{LGW})}, \mathcal{C}_{1,(b)}^{(\text{LGW})}, \mathcal{C}_{2,(b)}^{(\text{LGW})}\right)$ using the LGW-SI code construction in Subsection V-B1 for a LGW-SI setup with the following parameters:

- the encoder side-information alphabet $\mathcal{U}_0, \times \mathcal{U}_1 \times \mathcal{U}_2 \times \tilde{\mathcal{Y}}$;
- the decoder side-information alphabets $\mathcal{Y}_1$ and $\mathcal{Y}_2$;
- the reconstruction alphabets $\mathcal{V}_1$ and $\mathcal{V}_2$;
- the side-information law $P_{U_0,U_1,U_2,\tilde{Y},Y_1,Y_2}$; and
- the reconstruction laws $P_{V_1|U_0U_1U_2\tilde{Y}}(v_1|u_0,u_1,u_2,\tilde{y}) = \sum_{v_0,v_2} P_{V_0V_1V_2|U_0,U_1,U_2,\tilde{Y}}(v_0,v_1,v_2|u_0,u_1,u_2,\tilde{y})$ and $P_{V_2|U_0U_1U_2\tilde{Y}}(v_2|u_0,u_1,u_2,\tilde{y}) = \sum_{v_0,v_1} P_{V_0V_1V_2|U_0,U_1,U_2,\tilde{Y}}(v_0,v_1,v_2|u_0,u_1,u_2,\tilde{y})$.

As parameters of the LGW-SI code we choose

- the auxiliary alphabet $\mathcal{V}_0$;
- the conditional law $P_{V_0V_1V_2|U_0,U_1,U_2,\tilde{Y}}$;
- the nonnegative rates $\tilde{R}_0, \tilde{R}_1, \tilde{R}_2, \tilde{R}_0', \tilde{R}_1', \tilde{R}_2'$;
- the small number $\epsilon/2$; and
- the blocklength n.

[6]Without this assumption, the broadcast problem is not very interesting. It is easy to verify that when this assumption fails to hold, both our regions collapse to the trivial capacity region.

In what follows, let $\lambda_{(b)}$, $\Lambda_{1,(b)}$, and $\Lambda_{2,(b)}$ denote the LGW-SI encoding and decoding rules corresponding to these codes.

We are now ready to describe the scheme in detail.

1) Encoding: In the first block $b = 1$, the transmitter forms the indices $J_{0,(1)} = (M_{0,(1)}, 1)$, $J_{1,(1)} = (M_{1,(1)}, 1)$, and $J_{2,(1)} = (M_{2,(1)}, 1)$, and applies the encoding rule $\phi_{(1)}$ to this triplet $J_{0,(1)}, J_{1,(1)}, J_{2,(1)}$.

In blocks $b \in 2, \ldots, B$ the transmitter first applies the LGW-SI encoding function $\lambda_{(b-1)}$ to its "side-information" sequences $(U_{0,(b-1)}^n, U_{1,(b-1)}^n, U_{2,(b-1)}^n, \tilde{Y}_{(b-1)}^n)$ to generate the update messages $(K_{0,(b-1)}, K_{1,(b-1)}, K_{2,(b-1)})$. (Recall that by $U_{0,(b-1)}^n, U_{1,(b-1)}^n, U_{2,(b-1)}^n$ we denote the Marton auxiliary codewords produced in the previous encoding step.) The transmitter then generates the messages $J_{i,(b)} \stackrel{\text{def}}{=} (M_{i,(b)}, K_{i,(b-1)})$, and encodes them via the Marton-encoding rule $\varphi_{(b)}^{(n)}$. It finally sends the outcome of this encoding over the channel.

In the last block $B+1$, the transmitter first applies the LGW-SI encoding function $\lambda_{(B)}$ to the sequences $(U_{0,(B)}^n, U_{1,(B)}^n, U_{2,(B)}^n, \tilde{Y}_{(B)}^n)$ to generate the update messages $K_{i,(B)}$. It then forms the tuple $J_{0,(B+1)} = (1, K_{0,(B)})$, $J_{1,(B+1)} = (1, K_{1,(B)})$, and $J_{2,(B+1)} = (1, K_{2,(B)})$ and encodes them via the rule $\varphi_{(B+1)}$.

2) Decoding at Receiver i: Decoding is performed backwards, starting from the last block. The receiver first applies the decoding rule $\Phi_{i,(B+1)}$ to the outputs $Y_{i,(B+1)}^n$ attempting to decode the indices $(J_{0,(B+1)}, J_{i,(B+1)})$, and parses its guess $(\hat{J}_{0,(B+1)}, \hat{J}_{i,(B+1)})$ as $\hat{J}_{0,i,(B+1)} = (1, \hat{K}_{0,i,(B)})$ and $\hat{J}_{i,(B+1)} = (1, \hat{K}_{i,(B)})$.

Now, for every block $b \in \{2, \ldots, B\}$, starting with block B and going backwards, the receiver performs the following steps. It applies the LGW-SI decoder $\Lambda_{i,(b)}$ to its guess of the update messages $(\hat{K}_{0,i,(b)}, \hat{K}_{1,(b)})$ obtained in block $b+1$, and to its "side-information" $Y_{i,(b)}^n$. It then uses the produced guess $\hat{V}_{i,(b)}^n$ to decode the indices $(J_{0,(b)}, J_{i,(b)})$ sent in this block b. More precisely, it applies Marton's decoding rule $\Phi_{i,(b)}$ to the pair $(Y_{i,(b)}^n, \hat{V}_{i,(b)}^n)$ to decode the indices $(J_{0,(b)}, J_{i,(b)})$, and then parses its guess $(\hat{J}_{0,(b)}, \hat{J}_{i,(b)})$ as $\hat{J}_{0,i,(b)} = (\hat{M}_{0,i,(b)}, \hat{K}_{0,i,(b)})$ and $\hat{J}_{i,(b)} = (\hat{M}_{i,(b)}, \hat{K}_{i,(b)})$.

In the last processed block $b = 1$, the receiver applies the LGW-SI decoder $\Lambda_{i,(1)}$ to its guess of the update messages $(\hat{K}_{0,i,(1)}, \hat{K}_{1,(1)})$ produced in block 2 and to its "side-information" $Y_{i,(1)}^n$. It then applies Marton's decoding rule $\Phi_{i,(b)}$ to the pair $(Y_{i,(1)}^n, \hat{V}_{i,(1)}^n)$ to decode the indices $(J_{0,(1)}, J_{i,(1)})$ and parses its guess $(\hat{J}_{0,(1)}, \hat{J}_{i,(1)})$ as $\hat{J}_{0,i,(1)} = (\hat{M}_{0,i,(1)}, 1)$ and $\hat{J}_{i,(1)} = (\hat{M}_{i,(1)}, 1)$.

The receiver's guess of the messages M_0 and M_i is then given by the products $\hat{M}_{0,i} = (\hat{M}_{0,i,(1)}, \ldots, \hat{M}_{0,i,(B)})$ and $\hat{M}_i = (\hat{M}_{i,(1)}, \ldots, \hat{M}_{i,(B)})$.

3) Error Analysis: We bound the average probability of error averaged over the random messages, codes, and channel realizations. To this end, we define the block b "error" event $\mathcal{E}_b \stackrel{\text{def}}{=} \mathcal{E}_b^1 \cup \mathcal{E}_b^2$, where

$$\mathcal{E}_b^i \stackrel{\text{def}}{=} \left\{ (\widehat{J}_{0,i,(b)}, \widehat{J}_{i,(b)}) \neq (J_{0,(b)}, J_{i,(b)}) \right\}.$$

Let $\mathcal{E}^i$ be the event where Receiver i decodes (M_0, M_i) incorrectly, i.e.,

$$\mathcal{E}^i \stackrel{\text{def}}{=} \bigcup_{b=1}^{B} \left\{ (\widehat{M}_{0,i,(b)}, \widehat{M}_{i,(b)}) \neq \left(M_{0,(b)}, M_{i,(b)}\right) \right\}$$

and let $\mathcal{E} = \mathcal{E}^1 \cup \mathcal{E}^2$. The error probability for our scheme is given by $\Pr(\mathcal{E})$, which is upper bounded by

$$\Pr(\mathcal{E}) \leq \Pr(\mathcal{E}_{B+1}) + \sum_{b=1}^{B} \Pr(\mathcal{E}_b | \bar{\mathcal{E}}_{b+1}).$$

By construction, we have that $\Pr(\mathcal{E}_{B+1}) \to 0$ as $n \to \infty$. Let us now analyze the probability $\Pr(\mathcal{E}_b | \overline{\mathcal{E}}_{b+1})$ for a fixed $b \in [B]$. In light of Remark 2, we see that if

$$\begin{aligned}
\bar{R}_0 &< \min_i I(U_0; Y_i, V_i) - \delta(\epsilon) \\
\bar{R}_0 + \bar{R}_1 &< I(U_0, U_1; Y_1, V_1) - \delta(\epsilon) \\
\bar{R}_0 + \bar{R}_2 &< I(U_0, U_2; Y_2, V_2) - \delta(\epsilon) \\
\bar{R}_0 + \bar{R}_1 + \bar{R}_2 &< I(U_1; Y_1, V_1 | U_0) + I(U_2; Y_2, V_2 | U_0) \\
&\quad + \min_i I(U_0; Y_i, V_i) - I(U_1; U_2 | U_0) - \delta(\epsilon),
\end{aligned} \tag{39}$$

and

$$\Pr((U^n_{0,(b)}, U^n_{i,(b)}, Y^n_{i,(b)}, \hat{V}^n_{i,(b)}) \notin \mathcal{T}^n_\epsilon(P_{U_0 U_i Y_i, V_i})) \to 0 \tag{40}$$

as $n \to \infty$, then there exists a choice of the parameters such that $\Pr\left(E_b | \bar{E}_{b+1}\right) \to 0$ as $n \to \infty$. From this point forward we assume that conditions (39) hold. In the following we wish to prove that if additionally

$$\begin{aligned}
\widetilde{R}_1 &> I(U_0, U_1, U_2, \widetilde{Y}; V_1 | V_0, Y_1) + \delta(\epsilon) \\
\widetilde{R}_2 &> I(U_0, U_1, U_2, \widetilde{Y}; V_2 | V_0, Y_2) + \delta(\epsilon) \\
\widetilde{R}_0 + \widetilde{R}_1 &> I(U_0, U_1, U_2, \widetilde{Y}; V_0, V_1 | Y_1) + \delta(\epsilon) \\
\widetilde{R}_0 + \widetilde{R}_2 &> I(U_0, U_1, U_2, \widetilde{Y}; V_0, V_2 | Y_2) + \delta(\epsilon),
\end{aligned} \tag{41}$$

then the limit (40) holds. To this end, it is easy to see that

$$\begin{aligned}
&\Pr\Big((U^n_{0,(b)}, U^n_{i,(b)}, Y^n_{i,(b)}, \hat{V}^n_{i,(b)}) \notin \mathcal{T}^n_\epsilon(P_{U_0 U_i Y_i V_i})\Big) \\
&\leq \Pr\Big((U^n_{0,(b)}, U^n_{i,(b)}, Y^n_{i,(b)}, V^n_{i,(b)}) \notin \mathcal{T}^n_\epsilon(P_{U_0 U_i Y_i V_i})\Big) \\
&\quad + \Pr\Big(\widehat{V}^n_{i,(b)} \neq V^n_{i,(b)}\Big),
\end{aligned} \tag{42}$$

where $V^n_{1,(b)}$ and $V^n_{2,(b)}$ denote the codewords chosen by the LGW-encoding rule $\lambda_{(b)}$. Hence we shall now verify that under conditions (41), each of the terms on the right-hand-side of (42) vanishes as $n \to \infty$.

Since the input $X^n_{(b)}$ is a component-wise function of $(U^n_{0,(b)}, U^n_{1,(b)}, U^n_{2,(b)})$, the Markov condition $(\widetilde{Y}^n_{(b)}) \multimap X^n_{(b)} \multimap (U^n_{0,(b)}, U^n_{1,(b)}, U^n_{2,(b)})$ holds, and the channel law is memoryless, we have that $(\widetilde{Y}^n_{(b)})$ is $P_{\widetilde{Y}|U_0 U_1 U_2}$-independent given $(U^n_{0,(b)}, U^n_{1,(b)}, U^n_{2,(b)})$. Furthermore, from Marton's code construction and in light of Remark 1, we have that under conditions (39)

$$\Pr\left((U^n_{0,(b)}, U^n_{1,(b)}, U^n_{2,(b)}) \notin \mathcal{T}_{\epsilon/32}(P_{U_0 U_1 U_2})\right) \to 0.$$

Therefore, by the conditional typicality Lemma we obtain

$$\Pr\left((U^n_{0,(b)}, U^n_{1,(b)}, U^n_{2,(b)}, \widetilde{Y}^n_{(b)}) \notin \mathcal{T}_{\epsilon/16}(P_{U_0 U_1 U_2 \widetilde{Y}})\right) \to 0$$

as $n \to \infty$. Moreover, by similar arguments, $(Y^n_{1,(b)}, Y^n_{2,(b)})$ is $P_{Y_1 Y_2 | U_0 U_1 U_2 \widetilde{Y}}$-independent given $U^n_{0,(b)}, U^n_{1,(b)}, U^n_{2,(b)}, \widetilde{Y}^n_{(b)}$. Therefore, the conditions of the modified LGW-SI setup of Remark 4 are satisfied (recall we have used $\epsilon/2$ for the LGW-SI code). Therefore, by that same remark and under conditions (41) it holds that

$$\Pr\Big(\widehat{V}^n_{i,(b)} \neq V^n_{i,(b)}\Big) \to 0 \tag{43}$$

$$\Pr\Big((U^n_{0,(b)}, U^n_{1,(b)}, U^n_{2,(b)}, \widetilde{Y}^n_{(b)}, V^n_{i,(b)}) \notin \mathcal{T}^n_{\epsilon/2}(P_{U_0 U_1 U_2 \widetilde{Y} V_i})\Big) \to 0 \tag{44}$$

as $n \to \infty$. We note that $Y^n_{i,(b)}$ is $P_{Y i | U_0 U_1 U_2 \widetilde{Y}}$-independent given $(U^n_{0,(b)}, U^n_{1,(b)}, U^n_{2,(b)}, \widetilde{Y}^n_{(b)})$, and the Markov condition

$$V^n_{i,(b)} \multimap (U^n_{0,(b)}, U^n_{1,(b)}, U^n_{2,(b)}, \widetilde{Y}^n_{(b)}) \multimap Y^n_{i,(b)}$$

holds. Together with (44) and the conditional typicality Lemma, this implies that

$$\Pr\Big((U^n_{0,(b)}, U^n_{i,(b)}, V^n_{i,(b)}, Y^n_{i,(b)}) \notin \mathcal{T}^n_\epsilon(P_{U_0 U_i Y_i V_i})\Big) \to 0 \tag{45}$$

Combining (42), (43) and (45), the limit in (40) is established.

Combining Constraints (39) and (41), replacing $\bar{R}_i$ by $R_i + \tilde{R}_i$, and employing the Fourier-Motzkin elimination algorithm, we obtain that under the set of constraints (46) shown on top of the next page there exists a choice of the parameters such that the probability of error of our scheme tends to 0 as $n \to \infty$, for any ϵ small enough. Taking B to be arbitrarily large, any rate triplet satisfying the constraints (46) is achievable by our scheme. Simple manipulations yield (37). The existence of a deterministic coding scheme achieving the same region follows from standard arguments.

Let us now establish the inclusion $\mathcal{R}^*_{\text{inner}} \subseteq \mathcal{R}_{\text{inner}}$. We use the exact same construction, but with the region $\mathcal{R}^{\text{inner}*}_{\text{LGW}}$ in lieu of the region $\mathcal{R}^{\text{inner}}_{\text{LGW}}$ for the rate analysis. It is readily verified that the constrains set (46) is replaced by the constraint set (47) on the top of the next page. Now, let us restrict the choice of (V_0, V_1, V_2) such that it satisfies the more stringent Markov condition[7] $(V_0, V_1, V_2) \multimap (X, \widetilde{Y}) \multimap (Y_1, Y_2, U_0, U_1, U_2)$. It is easily verified that for such a choice, (U_0, U_1, U_2) can be substituted by X in all the LGW-SI expressions in (47), which results in (38). Clearly, by the LGW-SI regions inclusion, the region that corresponds to the constraints (47) is smaller than that corresponding to (46), hence $\mathcal{R}^*_{\text{inner}} \subseteq \mathcal{R}_{\text{inner}}$.

VII. Examples

A. The Generalized Dueck DMBC

Consider a generalized version of Dueck's DMBC [2] where all three binary channels are (possibly) noisy, and the first and third channels are corrupted by (possibly) different noises. The channel input consists of three bits, $X = (X_1, X_0, X_2)$, and each output of two bits, $Y_1 = (Y_{1,1}, Y_{1,0})$ and $Y_2 = (Y_{2,0}, Y_{2,2})$. The channel law $P_{Y_1 Y_2 | X}$ is described by the relations:

$$\begin{aligned}
Y_{1,0} = Y_{2,0} &= X_0 \oplus Z_0, \\
Y_{1,1} = X_1 \oplus Z_1, &\quad \text{and} \quad Y_{2,2} = X_2 \oplus Z_2,
\end{aligned}$$

[7] recall that X is a function of (U_0, U_1, U_2)

$$R_0 + R_1 < I(U_0, U_1; Y_1, V_1) - I(U_0, U_1, U_2, \widetilde{Y}; V_0, V_1|Y_1) \tag{46a}$$
$$R_0 + R_2 < I(U_0, U_2; Y_2, V_2) - I(U_0, U_1, U_2, \widetilde{Y}; V_0, V_2|Y_2) \tag{46b}$$
$$\begin{aligned} R_0 + R_1 + R_2 < {} & I(U_1; Y_1, V_1|U_0) + I(U_2; Y_2, V_2|U_0) + \min_i I(U_0; Y_i, V_i) - I(U_1; U_2|U_0) \\ & -I(U_0, U_1, U_2, \widetilde{Y}; V_2|V_0, Y_2) - I(U_0, U_1, U_2, \widetilde{Y}; V_0, V_1|Y_1) \end{aligned} \tag{46c}$$
$$\begin{aligned} R_0 + R_1 + R_2 < {} & I(U_1; Y_1, V_1|U_0) + I(U_2; Y_2, V_2|U_0) + \min_i I(U_0; Y_i, V_i) - I(U_1; U_2|U_0) \\ & -I(U_0, U_1, U_2, \widetilde{Y}; V_1|V_0, Y_1) - I(U_0, U_1, U_2, \widetilde{Y}; V_0, V_2|Y_2) \end{aligned} \tag{46d}$$
$$\begin{aligned} 2R_0 + R_1 + R_2 < {} & \min_i I(U_0; Y_i, V_i) + I(U_1; Y_1, V_1|U_0) + I(U_2; Y_2, V_2|U_0) + \min_i I(U_0; Y_i, V_i) \\ & -I(U_1; U_2|U_0) - I(U_0, U_1, U_2, \widetilde{Y}; V_0, V_1|Y_1) - I(U_0, U_1, U_2, \widetilde{Y}; V_0, V_2|Y_2) \end{aligned} \tag{46e}$$

$$R_0 < \min_{i\in\{1,2\}} I(U_0; Y_i, V_i) - \max_{i\in\{1,2\}} I(V_0; U_0, U_1, U_2, \widetilde{Y}|Y_i) \tag{47a}$$
$$R_0 + R_1 < I(U_0, U_1; Y_1, V_1) - I(U_0, U_1, U_2, \widetilde{Y}; V_1|V_0, Y_1) - \max_{i\in\{1,2\}} I(V_0; U_0, U_1, U_2, \widetilde{Y}|Y_i) \tag{47b}$$
$$R_0 + R_2 < I(U_0, U_2; Y_2, V_2) - I(U_0, U_1, U_2, \widetilde{Y}; V_2|V_0, Y_2) - \max_{i\in\{1,2\}} I(V_0; U_0, U_1, U_2, \widetilde{Y}|Y_i) \tag{47c}$$
$$R_0 + R_1 + R_2 < I(U_1; Y_1, V_1|U_0) + I(U_2; Y_2, V_2|U_0) + \min_{i\in\{1,2\}} I(U_0; Y_i, V_i) - I(U_1; U_2|U_0) \tag{47d}$$
$$-I(U_0, U_1, U_2, \widetilde{Y}; V_1|V_0, Y_1) - I(U_0, U_1, U_2, \widetilde{Y}; V_2|V_0, Y_2) \tag{47e}$$
$$- \max_{i\in\{1,2\}} I(V_0; U_0, U_1, U_2, \widetilde{Y}|Y_i) \tag{47f}$$

---

where Z_0, Z_1, Z_2 are binary random variables of a given law $P_{Z_0 Z_1 Z_2}$. We restrict attention to laws $P_{Z_0 Z_1 Z_2}$ such that

$$H(Z_0, Z_1) \le 1 \qquad \text{and} \qquad H(Z_0, Z_2) \le 1. \tag{48}$$

Under these conditions and when the transmitter only wishes to send private messages but no common message, our scheme achieves the noiseless-feedback capacity.

Theorem 4. *Under condition* (48) *and for* $R_0 = 0$*, the noiseless-feedback capacity of the Generalized Dueck DMBC is the set of all nonnegative rate pairs* (R_1, R_2) *satisfying*

$$\begin{aligned} R_1 &\le 2 - H(Z_0, Z_1), \\ R_2 &\le 2 - H(Z_0, Z_2), \\ R_1 + R_2 &\le 3 - H(Z_0, Z_1, Z_2). \end{aligned} \tag{49}$$

Proof. The converse follows from the cutset bound and from the fact that the no-feedback capacity of a DMBC depends on the channel law $P_{YZ|X}(y, z|x)$ only through the conditional marginal laws $P_{Y|X}(y|x) = \sum_z P_{YZ|X}(y, z|x)$ and $P_{Z|X}(z|x) = \sum_y P_{YZ|X}(y, z|x)$. The direct part follows from Theorem 3 by taking the convex hull of the two achievable regions that result when (37) is evaluated for the following two choices: (U_0, U_1, U_2) i.i.d. $\sim$ Bern(1/2); $X_0 = U_0$; $X_1 = U_1$; $X_2 = U_2$; $V_1 = (X_0, X_1)$; $V_2 = (X_0, X_2)$; and either $V_0 = (Z_0, Z_1)$ or $V_0 = (Z_0, Z_2)$. ∎

Corollary 1. *For zero common rate* $R_0 = 0$*, the no-feedback capacity of the generalized Dueck DMBC is the set of all nonnegative rate pairs* (R_1, R_2) *satisfying*

$$\begin{aligned} R_1 &\le 2 - H(Z_0, Z_1), \\ R_2 &\le 2 - H(Z_0, Z_2), \\ R_1 + R_2 &\le 3 - H(Z_0, Z_1, Z_2) - I(Z_1; Z_2|Z_0). \end{aligned}$$

Corollary 2. *Unless the triplet* $Z_1 - Z_0 - Z_2$ *forms a Markov chain, noiseless feedback strictly increases the capacity of the Generalized Dueck DMBC satisfying Conditions* (48)*.*

Let's briefly consider the case of noisy feedback $\widetilde{Y} = (Y_{1,1} \oplus W_1, Y_{1,0} \oplus W_0, Y_{2,2} \oplus W_2)$ where (W_0, W_1, W_2) are arbitrary distributed binary random variables, with marginals $W_i \sim \mathrm{Bern(q_i)}$. Evaluating Theorem 3 for this noisy-feedback setup is cumbersome and left out. But from Corollary 2 and the continuity considerations mentioned in Remark 8, we can conclude the following.

Corollary 3. *If* $Z_1 - Z_0 - Z_2$ *do not form a Markov chain, then for any sufficiently small value of* $\max\{q_0, q_1, q_2\}$*, the noisy feedback introduced above enlarges the capacity region of the Generalized Dueck DMBC.*

B. The Noisy Blackwell DMBC

Consider the following noisy version of the Blackwell DMBC [4]. The input alphabet is ternary $\mathcal{X} = \{0, 1, 2\}$ and both output alphabets are binary $\mathcal{Y}_1 = \mathcal{Y}_2 = \{0, 1\}$. Let $Z \sim$ Bern(p) for some $p < \frac{1}{2}$ and independent of X. The channel law $P_{Y_1 Y_2|X}$ is described as follows.

$$Y_1 = \begin{cases} Z & X = 0 \\ 1 - Z & X = 1, 2 \end{cases} \qquad Y_2 = \begin{cases} Z & X = 0, 1 \\ 1 - Z & X = 2 \end{cases}$$

Consider a noiseless feedback link $\widetilde{Y} = (Y_1, Y_2)$.

Let $U_0 \sim \text{Bern}(\frac{1}{2})$, and let (U_1, U_2) be jointly distributed binary random variables where $P_{U_1U_2}(\cdot,\cdot) = (\alpha, 1-\alpha-\beta, 0, \beta)$ (lexicographically ordered) if $U_0 = 0$, and $P_{U_1U_2}(\cdot,\cdot) = (\beta, 1-\alpha-\beta, 0, \alpha)$ otherwise, for some nonnegative α, β satisfying $\alpha + \beta \leq 1$. Set $X \stackrel{\text{def}}{=} U_1 + U_2$ (real addition).

Now, let $V_1 \stackrel{\text{def}}{=} U_1$ and $V_2 \stackrel{\text{def}}{=} U_2$. Note that this yields $Y_1 = V_1 \oplus Z$ and $Y_2 = V_2 \oplus Z$. Define further

$$V_0 \stackrel{\text{def}}{=} V_1 \oplus \widetilde{Y}_1 = Z$$

Note that the improved outputs here are simply the deterministic part of the channel actions on its input. Using Theorem 3 with (38) we get the following achievable region:

$$\begin{aligned}
R_0 &\leq h_b\left(\left(\frac{\alpha+\beta}{2}\right) \star p\right) - \frac{1}{2}(h_b(\alpha) + h_b(\beta)) - h_b(p) \\
R_0 + R_1 &\leq h_b\left(\left(\frac{\alpha+\beta}{2}\right) \star p\right) - h_b(p) \\
R_0 + R_2 &\leq h_b\left(\left(\frac{\alpha+\beta}{2}\right) \star p\right) - h_b(p) \\
R_0 + R_1 + R_2 &\leq h_b\left(\left(\frac{\alpha+\beta}{2}\right) \star p\right) + \frac{1-\beta}{2} h_b\left(\frac{\alpha}{1-\beta}\right) \\
&\quad + \frac{1-\alpha}{2} h_b\left(\frac{\beta}{1-\alpha}\right) - h_b(p)
\end{aligned}$$

Here $h_b(\cdot)$ is the binary entropy function. Let us consider the sum-rates $R_1 + R_2$ guaranteed by the region above. To that end, we set $R_0 = 0$ and note it is sufficient to consider only the last inequality[8]. We get:

$$\begin{aligned}
C_{\text{NoiselessFB},\Sigma} \geq \sup \Bigg\{ & h_b\left(\left(\frac{\alpha+\beta}{2}\right) \star p\right) \\
& + \frac{1-\beta}{2} h_b\left(\frac{\alpha}{1-\beta}\right) + \frac{1-\alpha}{2} h_b\left(\frac{\beta}{1-\alpha}\right) - h_b(p) \Bigg\}
\end{aligned} \tag{50}$$

where the supremum is taken over all nonnegative α, β with $\alpha + \beta \leq 1$, satisfying

$$h_b\left(\left(\frac{\alpha+\beta}{2}\right) \star p\right) - \frac{1}{2}(h_b(\alpha) + h_b(\beta)) - h_b(p) \geq 0$$

For comparison, let us now upper bound the corresponding no-feedback sum-capacity $\mathcal{C}_{\text{NoFB},\Sigma}$. Since the no-feedback capacity of a DMBC depends only on the marginals $P_{Y_1|X}, P_{Y_2|X}$, the capacity region for the Noisy Blackwell channel remains the same if we replace Z with Z_1, Z_2 for Y_1, Y_2 respectively, where $Z_i \sim \text{Bern}(p)$ are independent. Computing the cut-set upper bound for this setting, we get

$$\begin{aligned}
\mathcal{C}_{\text{NoFB},\Sigma} \leq \sup_{\alpha \in (0,\frac{1}{2})} \big\{ & H[(\alpha(p-\bar{p})^2 + p\bar{p}, \bar{p}^2 + 2\alpha p, \\
& p^2 + 2\alpha\bar{p}, \alpha(p-\bar{p})^2 + p\bar{p})] - 2h_b(p) \big\},
\end{aligned} \tag{51}$$

where $\bar{p} \stackrel{\text{def}}{=} 1 - p$. Figure 2 depicts the bounds (50) and (51) together with a cut-set upper bound on $C_{\text{NoiselessFB},\Sigma}$. Hence,

Corollary 4. *Noiseless feedback enlarges the capacity capacity region of the Noisy Blackwell-DMBC.*

[8]Note that $th_b(x/t) \leq h_b(x)$ for any $0 \leq x \leq t \leq 1$.

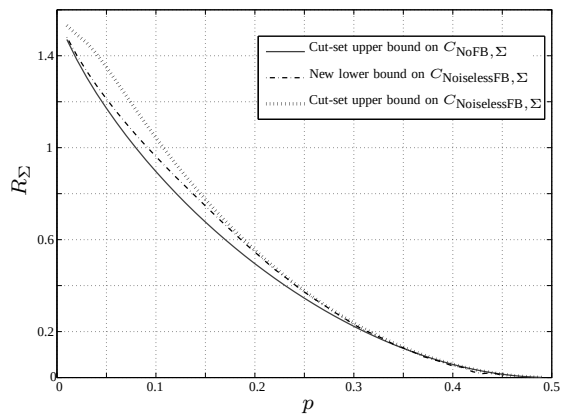

Fig. 2. Bounds on the sum-capacity of the noisy Blackwell DMBC

Corollary 5. *Let $\widehat{Y} = (Y_1 \oplus W_1, Y_2 \oplus W_2)$, where (W_1, W_2) are jointly distributed binary random variables with marginals $W_i \sim \text{Bern}(q_i)$, mutually independent of (X, Y_1, Y_2). By the continuity argument in Corollary 8, for any $p \in (0,1)$ and small enough $\max\{q_1, q_1\}$, noisy feedback strictly enlarges the capacity region of the Noisy Blackwell-DMBC with noisy feedback.*

Appendix A
Proof of Inequality (29)

Our proof is similar to a proof in [9, Appendix 12A]. Recall that we wish to prove:

$$\begin{aligned}
&\Pr\left(\bigcup_{\substack{\ell_i \in [2^{nR_i'}] \\ \ell_i \neq L_i^*}} (V_0^n(K_0^*, L_0^*), V_i^n(K_i^*, \ell_i), Y_i^n) \in T_\epsilon^n(P_{V_0V_iY_i})\right) \\
&\leq \Pr\left(\bigcup_{\ell_i=1}^{\lfloor 2^{nR_i'} \rfloor} (V_0^n(K_0^*, L_0^*), V_i^n(1, \ell_i), Y_i^n) \in T_\epsilon^n(P_{V_0V_iY_i})\right).
\end{aligned} \tag{52}$$

By the symmetry of the code construction, the probability that the codewords $\{V_i^n(K_i^*, \ell_i)\}_{\ell_i \neq L_i^*}$ are jointly typical with the pair $(V_0^n(K_0^*, L_0^*), Y_i^n)$ does not depend on the specific value of K_i^*, L_i^*. Therefore, Equality (53) on the next page holds. In the following we show Inequality (54), which combined with (53) establishes the desired inequality (52)

To this end we shall prove that Inequality (55) holds for all pairs $(k_i^*, \ell_i^*) \in [2^{nR_i}] \times [2^{nR_i'}]$. Since by the symmetry of our construction the pair (K_i^*, L_i^*) is independent and uniformly distributed, this proves the desired claim (54) as is shown in (56). We thus prove (55). To this end, we first notice that by the symmetry of the code construction, when $k_i^* = 1$, then the right-hand side of (55) does not depend on the value of ℓ_i^*. For the same reason, the right-hand side of (55) is the same for all pairs (k_i^*, ℓ_i^*) where $k_i^* \neq 1$. Thus, it suffices to prove (55) for the pairs $(k_i^*, \ell_i^*) = (1,1)$ and $(k_i^*, \ell_i^*) = (2,1)$. That (55) holds for $(k_i^*, \ell_i^*) = (1,1)$ is trivial. To prove (55) for $(k_i^*, \ell_i^*) = (2,1)$, we again use the symmetry of the code

$$\Pr\left(\bigcup_{\substack{\ell_i\in[2^{nR_i'}]\\ \ell_i\neq L_i^*}}(V_0^n(K_0^*,L_0^*),V_i^n(K_i^*,\ell_i),Y_i^n)\in T_\epsilon^n(P_{V_0V_iY_i})\right)$$
$$=\Pr\left(\bigcup_{\ell_i=2}^{\lfloor 2^{nR_i'}\rfloor}(V_0^n(K_0^*,L_0^*),V_i^n(1,\ell_i),Y_i^n)\in T_\epsilon^n(P_{V_0V_iY_i})\middle|K_i^*=1,L_i^*=1\right) \tag{53}$$

$$\Pr\left(\bigcup_{\ell_i=2}^{\lceil 2^{nR_i'}\rceil}(V_0^n(K_0^*,L_0^*),V_i^n(1,\ell_i),Y_i^n)\in T_\epsilon^n(P_{V_0V_iY_i})\middle|K_i^*=1,L_i^*=1\right)$$
$$\leq\Pr\left(\bigcup_{\ell_i=1}^{\lfloor 2^{nR_i'}\rfloor}(V_0^n(K_0^*,L_0^*),V_i^n(1,\ell_i),Y_i^n)\in T_\epsilon^n(P_{V_0V_iY_i})\right) \tag{54}$$

$$\Pr\left(\bigcup_{\ell_i=2}^{\lfloor 2^{nR_i'}\rfloor}(V_0^n(K_0^*,L_0^*),V_i^n(1,\ell_i),Y_i^n)\in T_\epsilon^n(P_{V_0V_iY_i})\middle|K_i^*=1,L_i^*=1\right)$$
$$\leq\Pr\left(\bigcup_{\ell_i=1}^{\lfloor 2^{nR_i'}\rfloor}(V_0^n(K_0^*,L_0^*),V_i^n(1,\ell_i),Y_i^n)\in T_\epsilon^n(P_{V_0V_iY_i})\middle|K_i^*=k_i^*,L_i^*=\ell_i^*\right) \tag{55}$$

$$\Pr\left(\bigcup_{\ell_i=2}^{\lfloor 2^{nR_i'}\rfloor}(V_0^n(K_0^*,L_0^*),V_i^n(1,\ell_i),Y_i^n)\in T_\epsilon^n(P_{V_0V_iY_i})\middle|K_i^*=L_i^*=1\right)$$
$$\leq\sum_{k_i^*\in[2^{nR_i}],\ \ell_i^*\in[2^{nR_i'}]}\frac{1}{\lfloor 2^{nR_i}\rfloor\lfloor 2^{nR_i'}\rfloor}\Pr\left(\bigcup_{\ell_i=1}^{\lfloor 2^{nR_i'}\rfloor}(V_0^n(K_0^*,L_0^*),V_i^n(1,\ell_i),Y_i^n)\in T_\epsilon^n(P_{V_0V_iY_i})\middle|K_i^*=k_i^*,L_i^*=\ell_i^*\right)$$
$$=\Pr\left(\bigcup_{\ell_i=1}^{\lfloor 2^{nR_i'}\rfloor}(V_0^n(K_0^*,L_0^*),V_i^n(1,\ell_i),Y_i^n)\in T_\epsilon^n(P_{V_0V_iY_i})\right). \tag{56}$$

$$\Pr\left(\bigcup_{\ell_i=2}^{\lfloor 2^{nR_i'}\rfloor}(V_0^n(K_0^*,L_0^*),V_i^n(2,\ell_i),Y_i^n)\in T_\epsilon^n(P_{V_0V_iY_i})\middle|K_i^*=1,L_i^*=1\right)$$
$$=\Pr\left(\bigcup_{\ell_i=2}^{\lfloor 2^{nR_i'}\rfloor}(V_0^n(1,1),V_i^n(1,\ell_i),Y_i^n)\in T_\epsilon^n(P_{V_0V_iY_i})\middle|K_i^*=2,L_i^*=1\right) \tag{57}$$

$$\Pr\left(\bigcup_{\ell_i=2}^{\lfloor 2^{nR_i'}\rfloor}(V_0^n(K_0^*,L_0^*),V_i^n(1,\ell_i),Y_i^n)\in T_\epsilon^n(P_{V_0V_iY_i})\middle|K_i^*=1,L_i^*=1\right)$$
$$=\Pr\left(\bigcup_{\ell_i=2}^{\lfloor 2^{nR_i'}\rfloor}(V_0^n(1,1),V_i^n(2,\ell_i),Y_i^n)\in T_\epsilon^n(P_{V_0V_iY_i})\middle|K_i^*=1,L_i^*=1\right) \tag{58}$$

construction to conclude Equality (57). Moreover, because conditioned on $K_i^* = 1$ and $L_i^* = 1$ every set of $\lfloor 2^{nR_i'} \rfloor - 1$ codewords $\{V_i^n(k_i, \ell_i)\}$ for $(k_i, \ell_i) \neq (1,1)$ has the same joint distribution, Inequality (58) on top of the next page holds. Combining finally (57) and (58) we obtain upper bound (55) for the desired pair $(k_i^* = 2, \ell_i^* = 1)$. This concludes our proof.

Appendix B
Convexity in Theorem 2

Let us show that $\mathcal{R}_{\text{LGW}}^{\text{inner}}$ is convex. Let $\{V_{0,j}, V_{1,j}, V_{2,j}, X_j, Y_{1,j}, Y_{2,j}\}_{j\in\{0,1\}}$ be two sets of mutually independent random variables for $j \in \{1,2\}$, where

- $(X_j, Y_{1,j}, Y_{2,j}) \sim P_{XY_1Y_2}$;
- $(V_{0,j}, V_{1,j}, V_{2,j}) - \!\circ\!- X_j - \!\circ\!- (Y_{1,j}, Y_{2,j})$.
- $P_{V_{i,j}|X_j} = P_{V_i|X}$ for $i \in \{1,2\}$.

Let $Q \sim \text{Bern}(\alpha)$ be independent of the union of the two sets. Define $\bar{V}_0 \stackrel{\text{def}}{=} V_{0,Q}, \bar{V}_i \stackrel{\text{def}}{=} V_{i,Q}, \bar{X} \stackrel{\text{def}}{=} X_Q, \bar{Y}_i \stackrel{\text{def}}{=} Y_{i,Q}$. Notice that as the law of $(X_1, Y_{1,1}, Y_{2,1})$ and the law of $(X_2, Y_{1,2}, Y_{2,2})$ are the same, the "time-sharing" random variable Q is independent of the triplet $(\bar{X}, \bar{Y}_1, \bar{Y}_2)$. Therefore, and since by assumption

$$(\bar{V}_0, \bar{V}_1, \bar{V}_2) - \!\circ\!- (\bar{X}, Q) - \!\circ\!- (\bar{Y}_1, \bar{Y}_2),$$

then defining $\tilde{V}_0 \stackrel{\text{def}}{=} (Q, \bar{V}_0)$, the following Markov chain also holds:

$$(\tilde{V}_0, \bar{V}_1, \bar{V}_2) - \!\circ\!- \bar{X} - \!\circ\!- (\bar{Y}_1, \bar{Y}_2).$$

We further notice that for $i \in \{1,2\}$:

$$I(\bar{X}; \bar{V}_i | \bar{V}_0, \bar{Y}_i, Q) = I(\bar{X}; \bar{V}_i | \tilde{V}_0, \bar{Y}_i)$$

and

$$I(\bar{X}; \bar{V}_0 | \bar{Y}_i, Q) = I(\bar{X}; \bar{V}_0, Q | \bar{Y}_i) \geq I(\bar{X}; \tilde{V}_0 | \bar{Y}_i).$$

where the first equality holds because of the independence of Q and $(\bar{X}, \bar{Y}_i)$, Thus, we conclude that the region is convex.

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